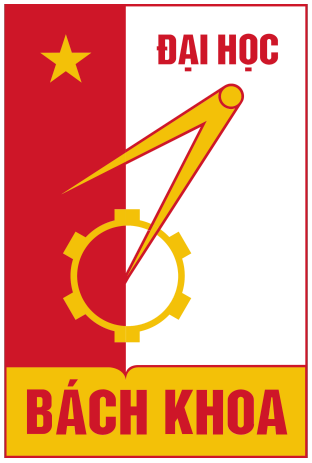
HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY
SCHOOL OF INFORMATION AND COMMUNICATION TECHNOLOGY



A Comparative Study of Statistical, Machine Learning, and Deep Learning Models for Multivariate Weather Forecasting

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Project Report

Course: IT2022E - Applied Statistics and Experimental Design

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Chapter 1

Introduction

1.1 Problem Motivation

Weather strongly influences agriculture, energy demand, transportation, public safety, and everyday decision making. Accurate short-term forecasts of variables such as temperature, humidity, and wind enable better planning and risk mitigation. While operational numerical weather prediction relies on solving the governing physical equations on supercomputers, data-driven models trained on historical observations offer a lightweight, station-level complement that can be deployed cheaply and retrained frequently. This project studies how well a range of statistical, machine-learning, and deep-learning models can forecast the next 24 hours of local weather from recent history.

1.2 Importance of Weather Forecasting

Reliable forecasting supports energy load balancing, renewable-generation scheduling, agricultural irrigation planning, and early warning for adverse conditions. Even modest reductions in forecast error translate into measurable operational and economic benefits.

1.3 Research Objectives

The objectives of this project are to:

1. Establish a single, shared experimental protocol (data handling, split, horizon, and metrics) so that heterogeneous models are compared fairly.
2. Implement and evaluate five forecasting models: DLinear, SARIMA, VAR, XGBoost, and LSTM.
3. Provide a detailed treatment of the XGBoost model, including feature engineering tailored to multivariate hourly time series.

4. Quantify and analyse forecast accuracy overall, per forecast horizon, and per target variable.

1.4 Project Scope

The study targets *short-term* forecasting: a fixed 24-hour horizon from a fixed look-back window, for a single location described by hourly observations. The seven forecast targets and the evaluation procedure are held constant across all models. Long-range (sub-seasonal/seasonal) forecasting, spatial fields, and ensemble physical models are out of scope.

1.5 Report Organisation

Chapter 2 describes the dataset. Chapter 3 presents an exploratory-data-analysis template. Chapter 4 details preprocessing. Chapter 5 presents the five forecasting models, with XGBoost covered in depth. Chapter 6 defines the evaluation protocol and metrics, then reports and discusses the results.

Chapter 2

Dataset Description

2.1 Data Source

The dataset is an hourly weather export in the Open-Meteo style [10], stored as `weather_dataset.csv`.

2.2 Data Collection Period

The data span **2021-01-01** to **2025-12-31** at hourly resolution, giving **43,824** rows with no gaps and no missing values.

2.3 Features Description

Of the raw hourly export, this study uses the columns in Table 2.1: five variables that are modelled directly plus wind speed and direction, which are combined into the Cartesian wind components `wx`/`wy` (Chapter 4). The raw file also contains precipitation, layered (low/mid/high) cloud cover, wind gusts, and topsoil moisture; the group agreed to exclude these from the seven-variable forecasting target set, so they are not used in any model and are not analysed further.

Table 2.1: Dataset columns used in this study and their clean names.

Raw column	**Clean name**	**Description**
`temperature_2m (°C)`	`temp`	Air temperature at 2 m (target)
`relative_humidity_2m (%)`	`rh`	Relative humidity at 2 m (target)
`dew_point_2m (°C)`	`dew`	Dew point at 2 m (target)
`surface_pressure (hPa)`	`pressure`	Surface pressure (target)
`cloud_cover (%)`	`cloud`	Total cloud cover (target)
`wind_speed_10m (km/h)`	`wind_speed`	Wind speed at 10 m (input for `wx`/`wy`)
`wind_direction_10m (°)`	`wind_dir`	Wind direction at 10 m (input for `wx`/`wy`)
`wx`, `wy` (km/h)		Cartesian wind components, engineered (targets)

2.4 Target Variable Description

All models forecast the same seven targets:

`temp`, `rh`, `dew`, `pressure`, `cloud`, `wx`, `wy`,

where `wx` and `wy` are the Cartesian wind components derived from wind speed and direction (Chapter 4). Each forecast is a 24×7 block (24 hours ahead $\times$ 7 variables).

2.5 Dataset Characteristics

The series is strictly hourly and complete, which simplifies windowing and avoids imputation in the common case.

Chapter 3

Exploratory Data Analysis (EDA)

This chapter analyses the cleaned dataset to characterise its distributions, dependencies, and temporal structure, and to motivate the preprocessing and modelling choices in later chapters. All figures are produced by `report/eda/run_eda.py`.

3.1 Dataset Overview

The dataset is complete and strictly hourly (Table 3.1): five full calendar years with no missing timestamps and no missing values across any of the 43,824 rows. A column-wise scan returns zero nulls and the hourly index is gap-free, so the time-interpolation step in Chapter 4 acts only as a safeguard (a missing-value heatmap is omitted, as it would be uniformly empty).

Table 3.1: Dataset overview.

Property	**Value**
Number of samples (rows)	43,824
Number of columns used	7 (5 direct + wind speed/direction)
Number of target variables	7 (`temp,rh,dew,pressure,cloud,wx,wy`)
Temporal resolution	Hourly
Data period	2021-01-01 – 2025-12-31
Missing values	0

3.2 Statistical Summary

Table 3.2 reports descriptive statistics for the seven targets. The profile is that of a warm, humid, monsoonal climate: a high mean temperature (24.0 °C), very high relative humidity (mean 78.8%, median 81%), persistently high cloud cover (mean 73.4%, median

99%), and surface pressure tightly concentrated around 1010 hPa (std 7.4). The wind components are centred near zero (`wx` mean 0.3, `wy` mean 3.8 km/h) with a wide spread.

Table 3.2: Summary statistics for the seven targets (full 2021–2025 period, native units).

Variable	Mean	Median	Std	Min	Max
`temp` (°C)	24.01	24.90	5.56	6.30	39.00
`rh` (%)	78.75	81.00	14.86	22.00	100.00
`dew` (°C)	19.81	22.10	6.36	-8.90	28.80
`pressure` (hPa)	1009.67	1009.30	7.41	982.00	1032.90
`cloud` (%)	73.41	99.00	37.26	0.00	100.00
`wx` (km/h)	0.28	0.00	8.16	-22.69	57.23
`wy` (km/h)	3.83	4.25	4.66	-36.69	45.02

3.3 Distribution Analysis

The marginal distributions (Fig. 3.1) and boxplots (Fig. 3.2) confirm several features that shape modelling: `cloud` is strongly bimodal, piling up at 0% and 100%; `rh`, `temp`, and `dew` are left-skewed toward their upper range; `pressure` is approximately symmetric; and the wind components `wx`, `wy` are roughly symmetric about zero with heavy tails. The boxplots show that the heavy "outliers" are concentrated in the wind components — genuine meteorological extremes (gusts/storms) rather than sensor errors — which supports keeping them and using an outlier-robust loss (MAE).

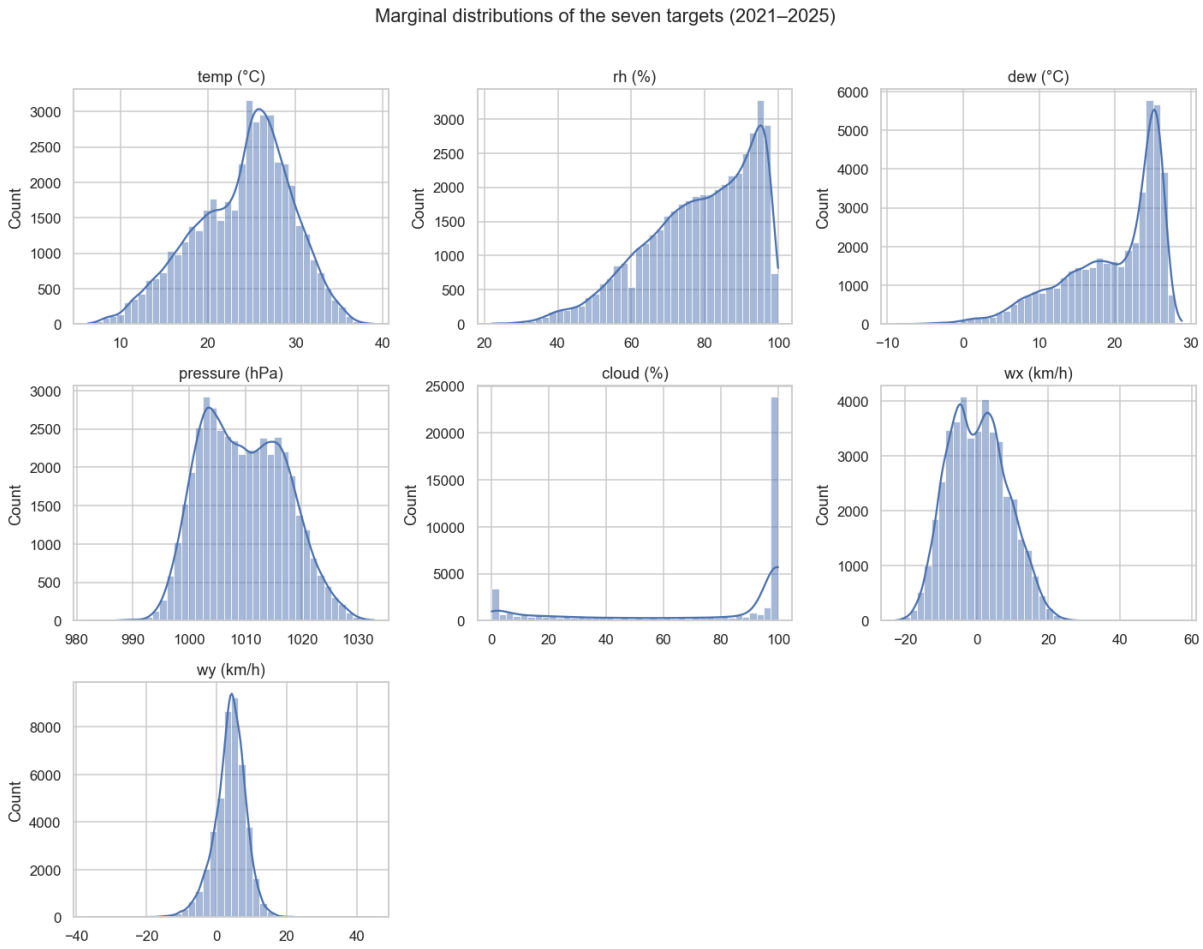


Figure 3.1: Marginal distributions of the seven targets.

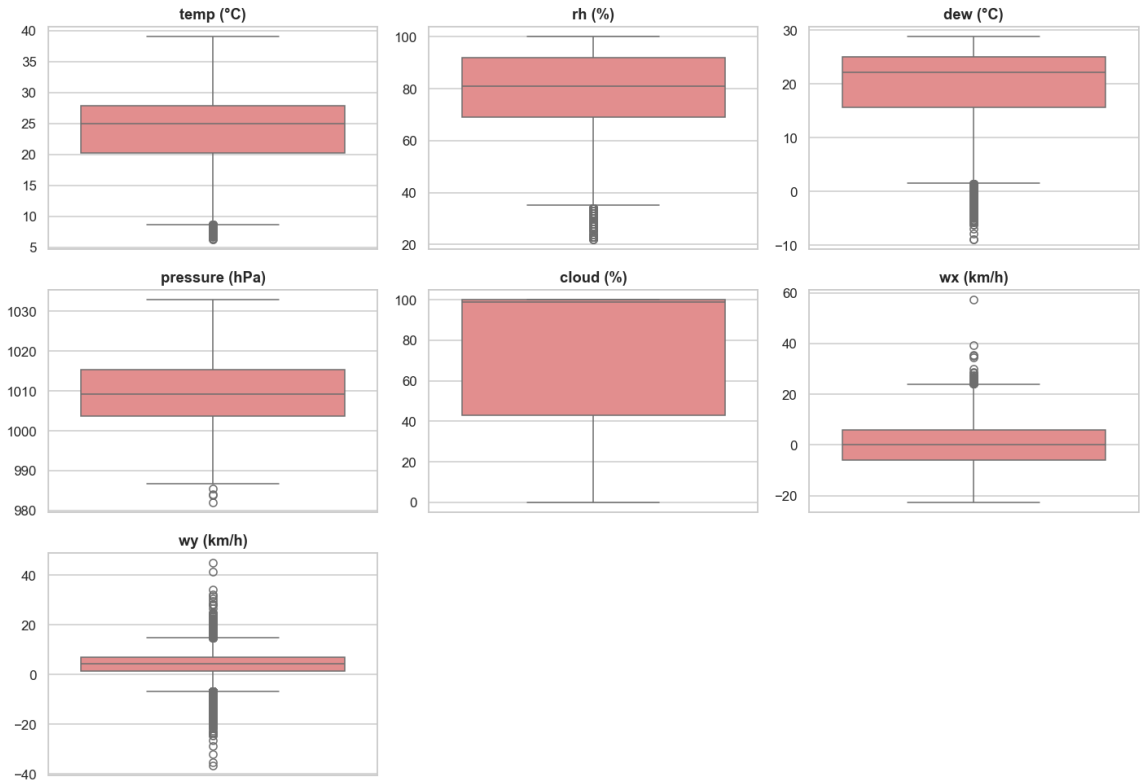


Figure 3.2: Per-variable boxplots of the seven targets for outlier inspection.

The wind rose (Fig. 3.3) shows the joint distribution of wind speed and direction. The wind is clearly *not* uniform over the compass — it concentrates in a few sectors, and the stronger speeds cluster there too. This matters for modelling because direction is a *circular* quantity ($0° \equiv 360°$): fed to a model as a raw angle, the wrap-around would be read as a huge jump. It is therefore decomposed into the Cartesian components $wx = s\cos\theta$ and $wy = s\sin\theta$ (Chapter 4), which the boxplots above show as the symmetric, heavy-tailed targets the models actually predict.

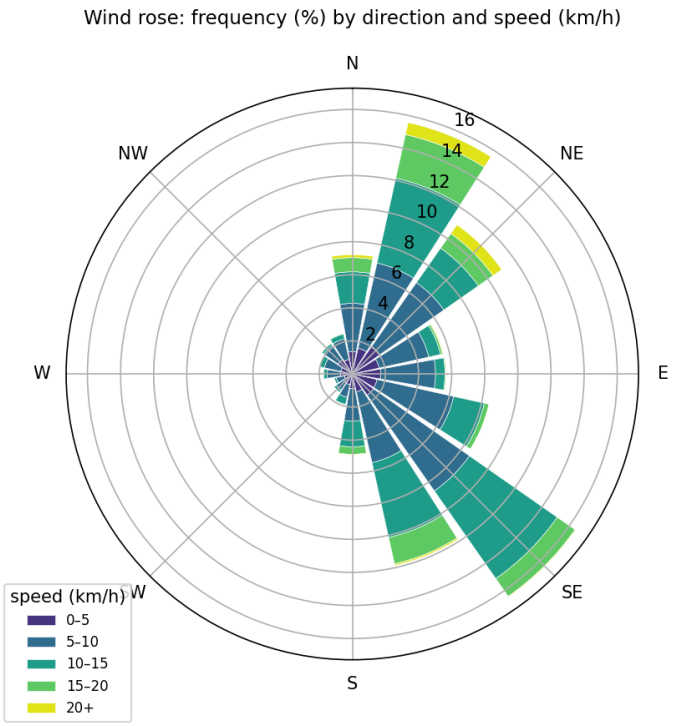


Figure 3.3: Wind rose: frequency of observations by direction (16 sectors), stacked by wind-speed band. Motivates decomposing the circular wind direction into the Cartesian targets `wx`, `wy`.

3.4 Correlation Analysis

The correlation matrix (Fig. 3.4) reveals strong, physically sensible dependencies: `temp`–`dew` (+0.85), `dew`–`pressure` (−0.87), and `temp`–`pressure` (−0.81). Relative humidity correlates moderately with dew point (+0.52), while `cloud` and the wind components are only weakly linearly related to the other targets — an early indication that `cloud` and wind will be the hardest to forecast. These inter-variable links justify a *multivariate* approach and the lag and rolling-window features used by the machine-learning models (Chapter 5).

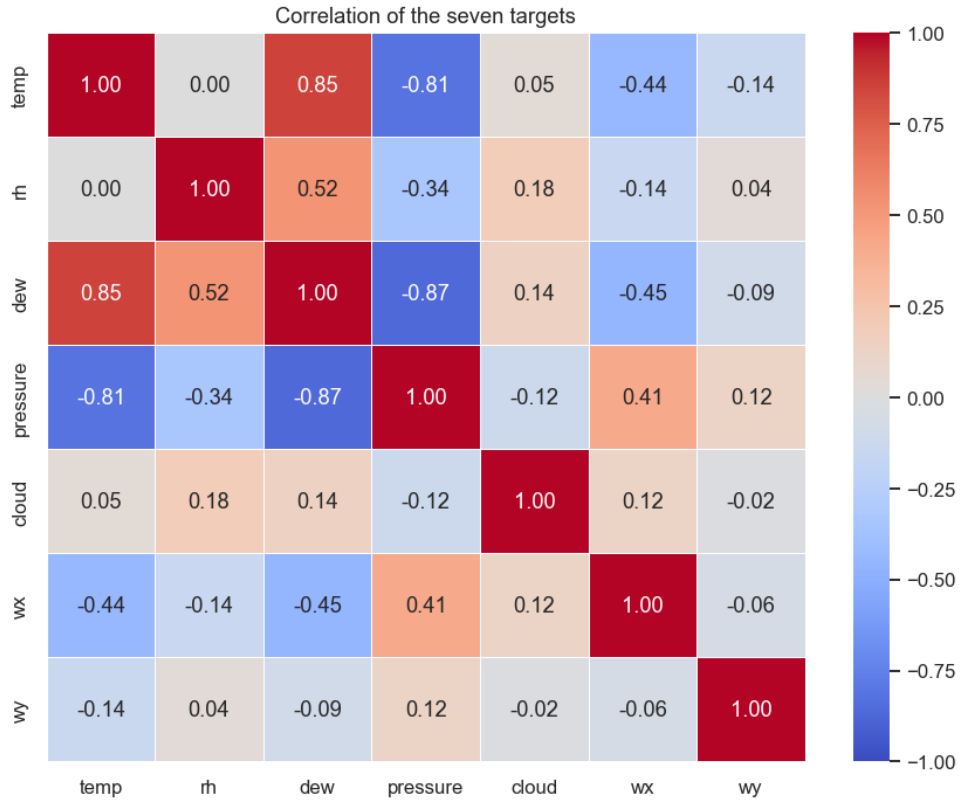


Figure 3.4: Correlation matrix of the seven targets.

3.5 Temporal Analysis

Because the task is a **24-hour-ahead hourly** forecast, the cycles that matter most are the ones that recur *within* that horizon. Three time scales are examined here: the diurnal cycle (dominant), the weekly scale (absent), and the annual cycle (context only).

Diurnal cycle (dominant). Averaging each target by hour of day (Fig. 3.5, one panel per target in real units) exposes the 24-hour cycle directly: `temp` peaks in the afternoon and troughs before dawn, `rh` moves in near-perfect anti-phase, the wind components show a clear daytime/nighttime pattern, and `cloud` and `pressure` have weaker, phase-shifted cycles. The afternoon temperature peak near 07–08h indicates the timestamps are in UTC for this location. This is the periodicity the models must capture within the forecast horizon.

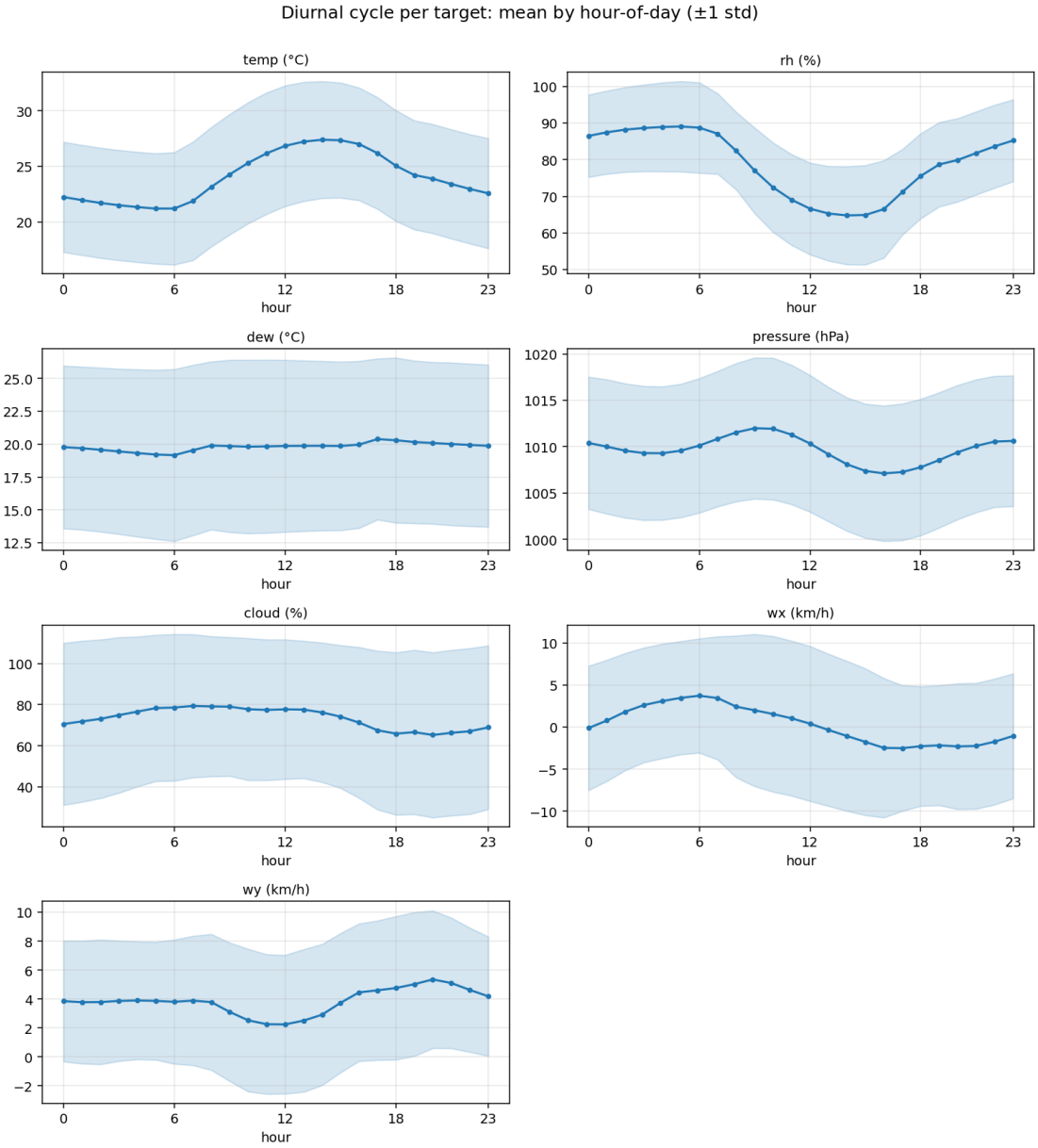


Figure 3.5: Diurnal cycle per target: mean by hour of day (±1 standard deviation), in real units.

Weekly scale (no seasonality). The day-of-week means (Fig. 3.6) are essentially flat — each target's weekday mean stays well inside its own daily spread — so weather has *no weekly seasonality* and no day-of-week feature is warranted, a useful negative result.

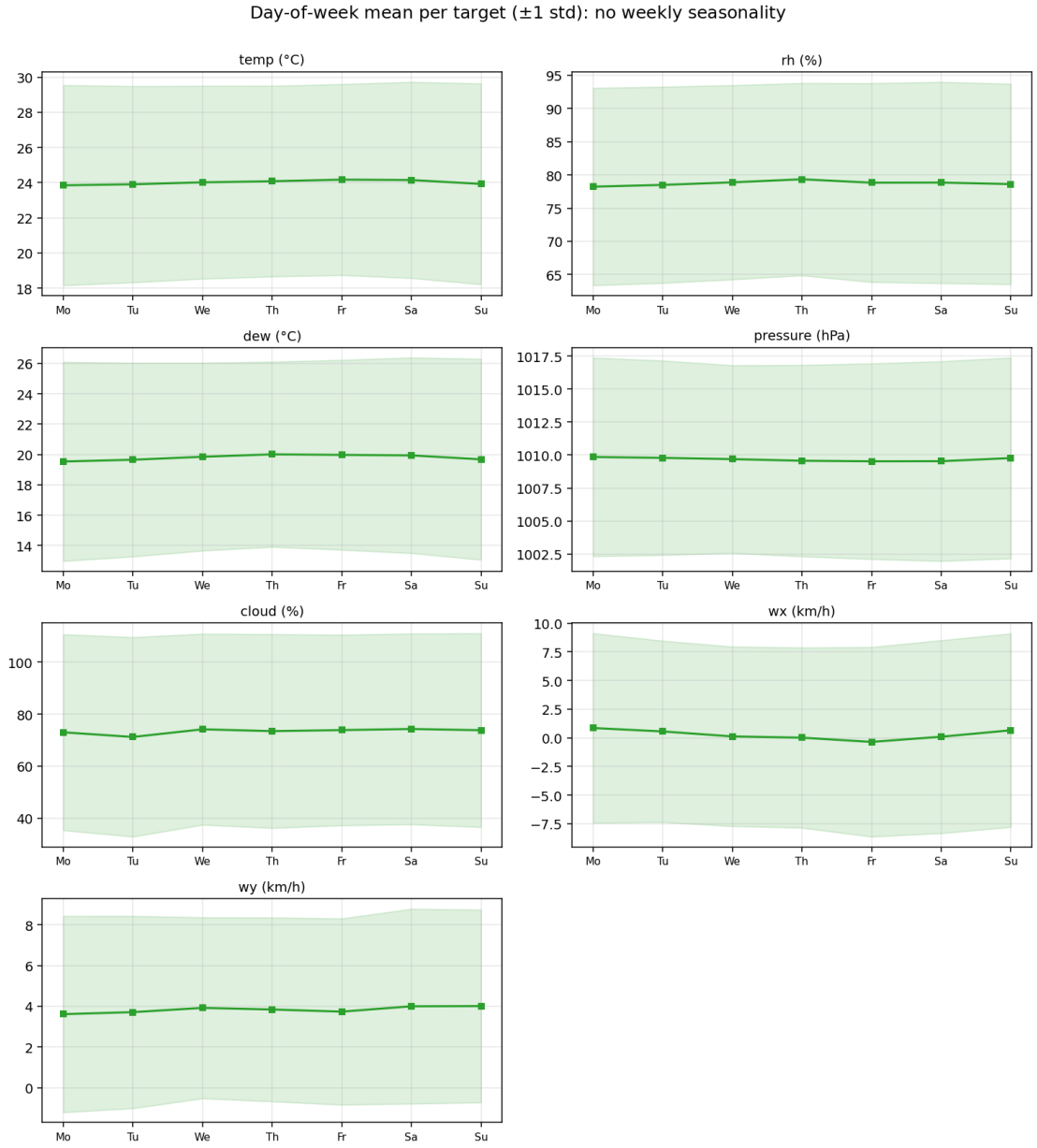


Figure 3.6: Day-of-week mean of each target (±1 standard deviation). Every profile is flat within its own spread — there is no weekly seasonality.

Annual cycle (context). The annual cycle is also present — weekly means (Fig. 3.7) show temperature and humidity swinging seasonally — but for a 24-hour forecast it acts as a slowly-varying *level* rather than a within-horizon period. It is captured by conditioning on recent history and by the day-of-year encoding, not by an explicit yearly seasonal term. In short, the **diurnal** cycle is the model-relevant periodicity; the annual cycle is context. Both motivate the cyclical hour-of-day and day-of-year encodings (Chapter 4).

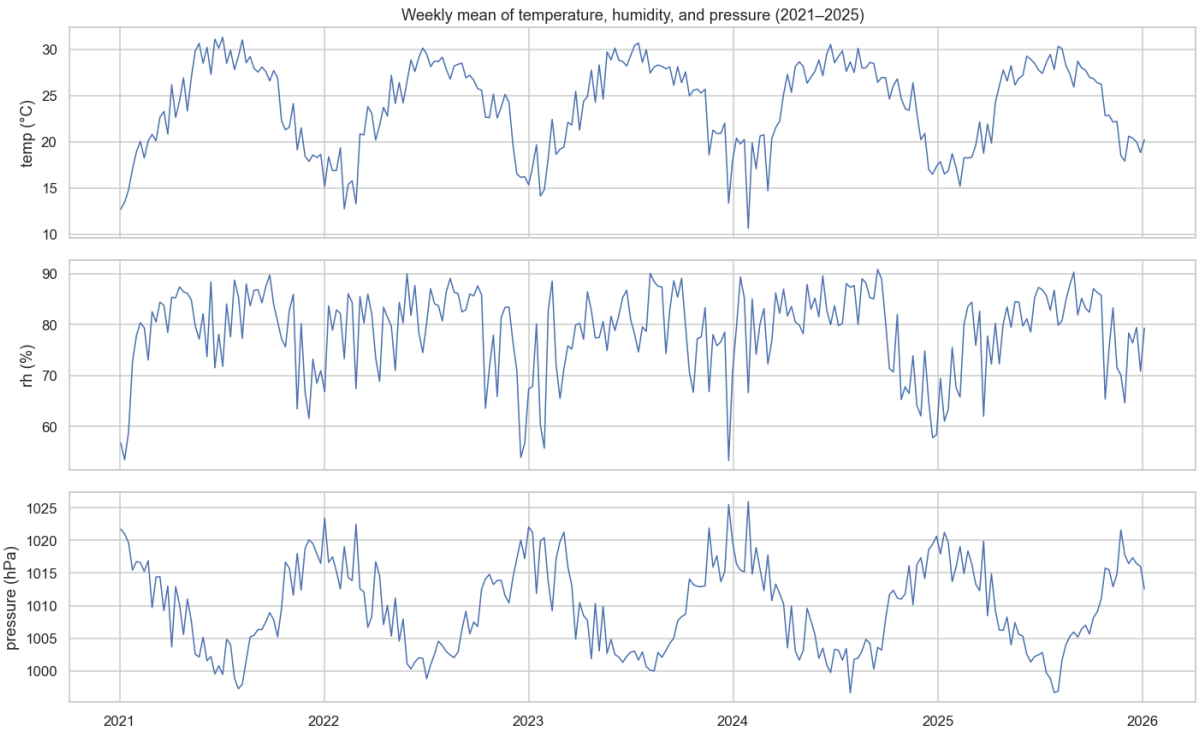


Figure 3.7: Weekly mean of temperature, humidity, and pressure over the full period (annual context).

Putting it together: trend / seasonal / residual. A period-24 additive decomposition of each target (Fig. 3.8) separates a multi-day *trend*, the repeating 24-h *seasonal* (diurnal) component, and a *residual*. Three points stand out:

- **The residual share predicts difficulty.** The residual holds only 0.5–4% of the variance for `temp`, `dew`, and `pressure`, but 37% for `cloud` and 42% for `wy` — the same ordering as the forecast error in Chapter 6. What the decomposition cannot explain, the models cannot either.
- **Predictability has two different sources.** `temp` and `rh` are carried by a strong diurnal-seasonal component (seasonal strength 0.80 and 0.72), whereas `dew` has almost no diurnal cycle (strength 0.13) yet is highly predictable through its slow *trend* (strength 0.98). The diurnal cycle is dominant but not universal — some targets are carried by persistence instead.
- **It validates the model designs.** Every series splits cleanly into trend + seasonal + (small or large) noise — precisely the structure the forecasting models build on (Chapter 5): a trend + seasonal decomposition, and an explicit 24-h seasonal component. For the weak-seasonal targets (`cloud`, `wy`) the trend and autoregressive parts must carry the forecast.

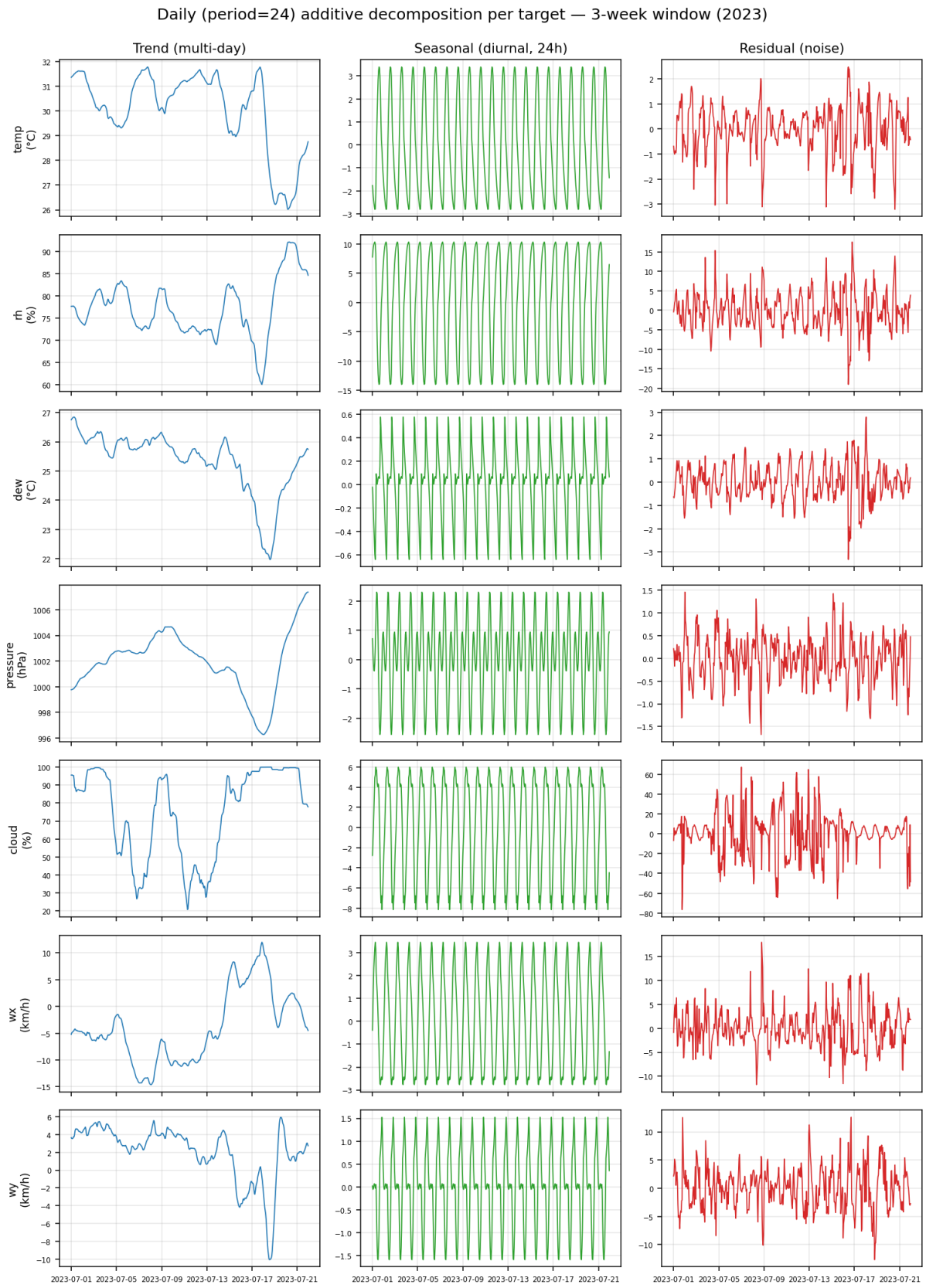


Figure 3.8: Additive decomposition (period = 24 h) of each target into trend, diurnal seasonal, and residual, over a 3-week window (2023). Note the small residual for `temp`/`dew`/`pressure` versus the large residual for `cloud` and `wy`, and that `dew`'s structure is almost all trend rather than diurnal.

3.6 Time-Series Structure

This section adds the time-series diagnostics that directly inform the statistical forecasting models introduced in Chapter 5: stationarity, autocorrelation structure, and lagged cross-dependence.

3.6.1 Stationarity

A series is (weakly) *stationary* if its mean, variance, and autocorrelation do not change over time; this is a precondition for the statistical forecasting models, which otherwise need differencing. Two complementary hypothesis tests are applied to each target, with deliberately *opposite* null hypotheses so that agreement between them is strong evidence.

Augmented Dickey–Fuller (ADF). The ADF test looks for a *unit root* — a stochastic trend that makes a series non-stationary. It fits the regression

$$\Delta y_t = \alpha + \beta t + \gamma\, y_{t-1} + \sum_{i=1}^{q} \delta_i\, \Delta y_{t-i} + \varepsilon_t, \tag{3.1}$$

where $\Delta y_t = y_t - y_{t-1}$ and the lagged differences Δy_{t-i} soak up short-run autocorrelation. The null is $H_0 : \gamma = 0$ (a unit root, *non*-stationary); the test statistic is the t-ratio $\hat{\gamma}/\,\mathrm{SE}(\hat{\gamma})$, compared against Dickey–Fuller (not Student-t) critical values. A *small* p-value rejects H_0 and so indicates stationarity.

KPSS. KPSS reverses the logic: its null is *stationarity*. Writing the partial sums of the regression residuals as $S_t = \sum_{i=1}^{t} \hat{e}_i$, the statistic is

$$\mathrm{KPSS} = \frac{1}{T^2} \sum_{t=1}^{T} \frac{S_t^2}{\hat{\sigma}_\infty^2}, \tag{3.2}$$

with $\hat{\sigma}_\infty^2$ a long-run variance estimate. A *large* statistic (small p) rejects stationarity. Because H_0 here is the opposite of ADF's, a *large* KPSS p-value supports stationarity.

How the test is run. Each target's full hourly series is passed to `statsmodels`' `adfuller` and `kpss`; we read off the two p-values and combine them: ADF small *and* KPSS large $\Rightarrow$ confidently stationary; the two disagreeing flags a borderline case (typically deterministic seasonality rather than a true unit root). The full formal statements are repeated for the models in Section 5.1.

The results (Table 3.3) show that ADF rejects the unit root for all seven targets at $p \le 10^{-12}$, and KPSS agrees (fails to reject stationarity) for five of them. For `temp` and `wx`, KPSS marginally rejects ($p = 0.049$ and 0.030); combined with the decisive ADF rejection, this points to *deterministic* annual seasonality rather than a stochastic trend —

a pattern also visible as the wavy-but-flat rolling mean in Fig. 3.9, whose rolling standard deviation stays roughly constant. Differencing is therefore not required, so the statistical models can be fit on the levels ($d = 0$).

Table 3.3: Stationarity tests on the seven targets (full 2021–2025 series).

Variable	**ADF** p	**KPSS** p	**Verdict**
`temp`	6.6×10^{-13}	0.049	stationary (borderline KPSS: annual cycle)
`rh`	4.3×10^{-29}	≥ 0.10	stationary
`dew`	2.1×10^{-14}	≥ 0.10	stationary
`pressure`	6.5×10^{-13}	≥ 0.10	stationary
`cloud`	$< 10^{-30}$	≥ 0.10	stationary
`wx`	2.9×10^{-30}	0.030	stationary (borderline KPSS: annual cycle)
`wy`	$< 10^{-30}$	≥ 0.10	stationary

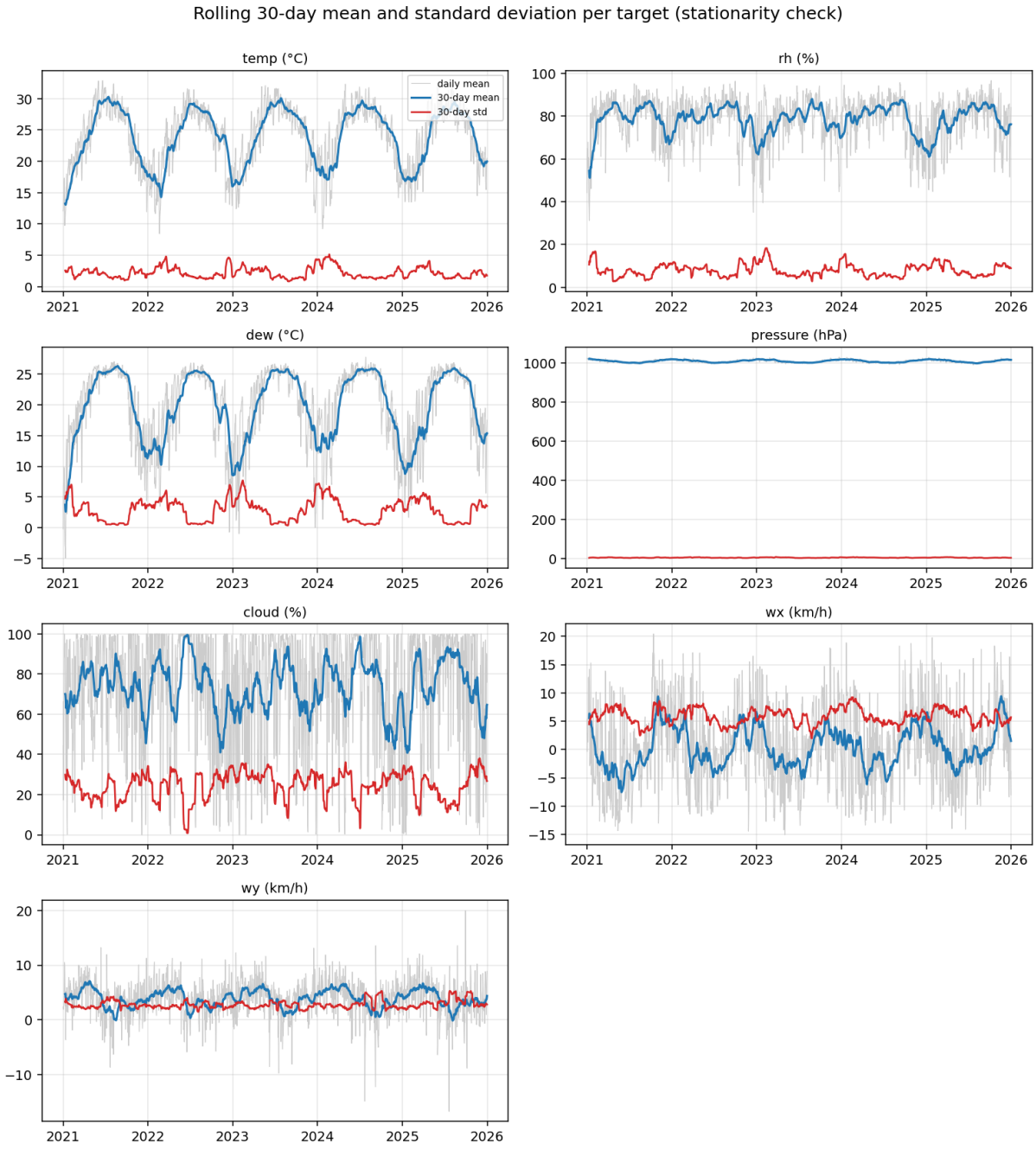


Figure 3.9: Rolling 30-day mean (blue) and standard deviation (red) of each target's daily series, in real units. The mean is flat apart from an annual wave (seasonality, not a unit-root trend) and the standard deviation is roughly constant — consistent with stationarity.

3.6.2 Autocorrelation Structure

The autocorrelation (ACF) and partial-autocorrelation (PACF) functions (Fig. 3.10) characterise each target's memory, and they fall into *two regimes*. The strongly diurnal targets (`temp`, `rh`, and to a lesser extent the wind components) have an *oscillating* ACF: it dips near lag 12 (the half-day anti-phase) and recovers at lag 24 (e.g. `temp` falls to 0.64 at 12 h and rises back to 0.93 at 24 h), the classic daily-cycle signature, with a PACF that cuts

off after a few lags plus a spike near lag 24. The slowly-varying targets `dew` and `pressure` instead have an ACF that stays near 1 and decays *monotonically* with almost no daily wiggle (≈ 0.95 at 12 h, ≈ 0.93 at 24 h, still ≈ 0.84 at two days): they are long-memory, synoptic-scale series with little diurnal content, exactly as the decomposition shows (`dew` trend-strength 0.98 but diurnal-strength only 0.13). Either way the structure is autoregressive — a low base order, augmented by daily-cycle lags for the first group and by long persistence for the second — which the autoregressive models later pick up through their lag-order selection (Chapter 5).

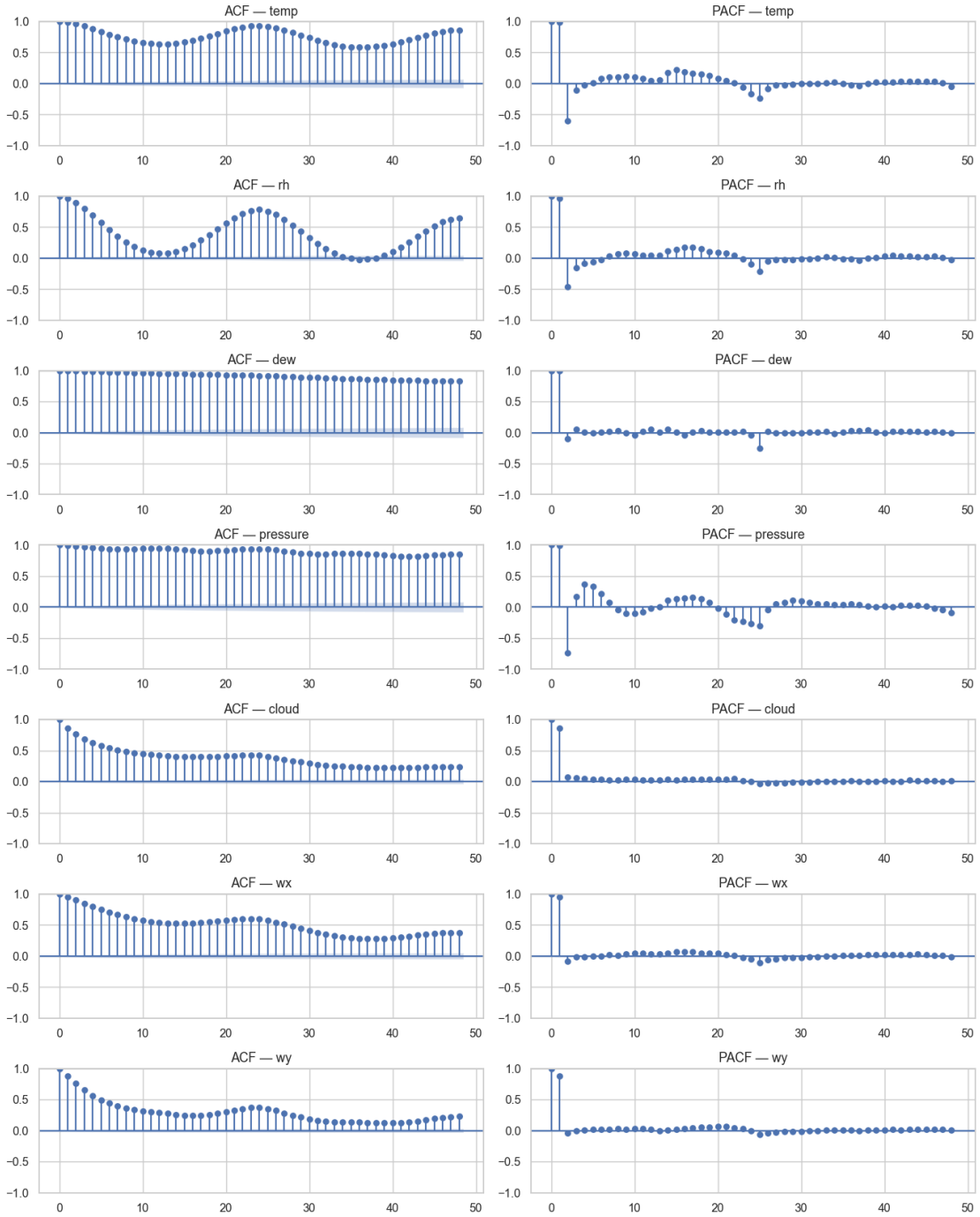


Figure 3.10: ACF (left) and PACF (right) of each target, lags 0–48 hours.

3.6.3 Lagged Cross-Correlation

Finally, the lagged cross-correlation matrix (Fig. 3.11) measures how each target's recent past ($t - 3\,\mathrm{h}$) relates to each other target's present. Strong off-diagonal entries appear among the thermodynamic variables — `dew`–`pressure` (-0.87), `temp`–`dew` ($+0.85$), and `temp`–`pressure` (-0.84) — meaning one variable's history carries predictive information about another. This lagged cross-dependence is exactly the signal a *multivariate* model can exploit but seven independent univariate models cannot, and it motivates the multivariate approach taken in Chapter 5.

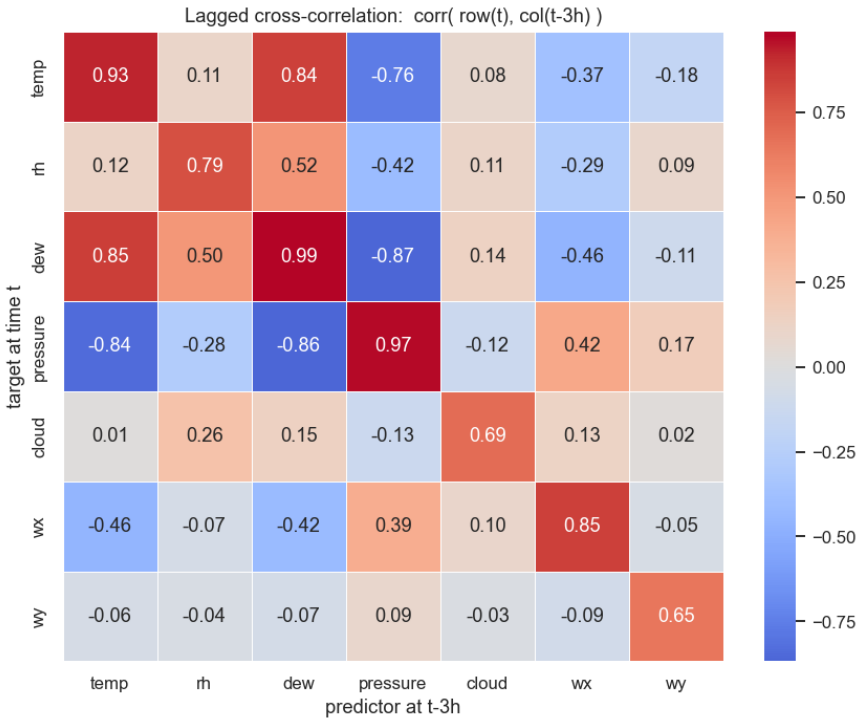


Figure 3.11: Lagged cross-correlation $\mathrm{corr}\big(\mathrm{row}(t),\, \mathrm{col}(t-3\mathrm{h})\big)$ among the seven targets.

That matrix is a snapshot at a single lag; tracing the cross-correlation *function* across $\pm 24\,\mathrm{h}$ (Fig. 3.12) shows how the dependence is distributed over lags and, crucially, its *direction*:

- **Lead–lag among the thermodynamic variables.** `temp`–`dew` peaks at lag 0 (0.85, essentially contemporaneous), but `temp` *leads* `pressure` by about 3 h (-0.84) and `pressure` in turn *leads* `dew` by about 3 h (-0.87) — a short causal chain a multivariate model can ride.

- **Structure a contemporaneous matrix misses entirely.** `temp` and `rh` are *uncorrelated at lag* 0 ($r \approx 0.00$) yet reach $r \approx 0.53$ at a 12 h offset — the half-day diurnal anti-phase. A same-time correlation matrix would call them unrelated; the lagged view shows a strong, exploitable relationship.

- **The dependence is broad in lag.** The thermodynamic curves stay high out to ±12–24 h, which is direct evidence for a relatively long lag order in the autoregressive models rather than a one- or two-lag model. By contrast `wx`–`wy` and `temp`–`cloud` stay weak ($|r| \leq 0.14$) at every lag, consistent with cloud and wind being the hardest, least cross-predictable targets.

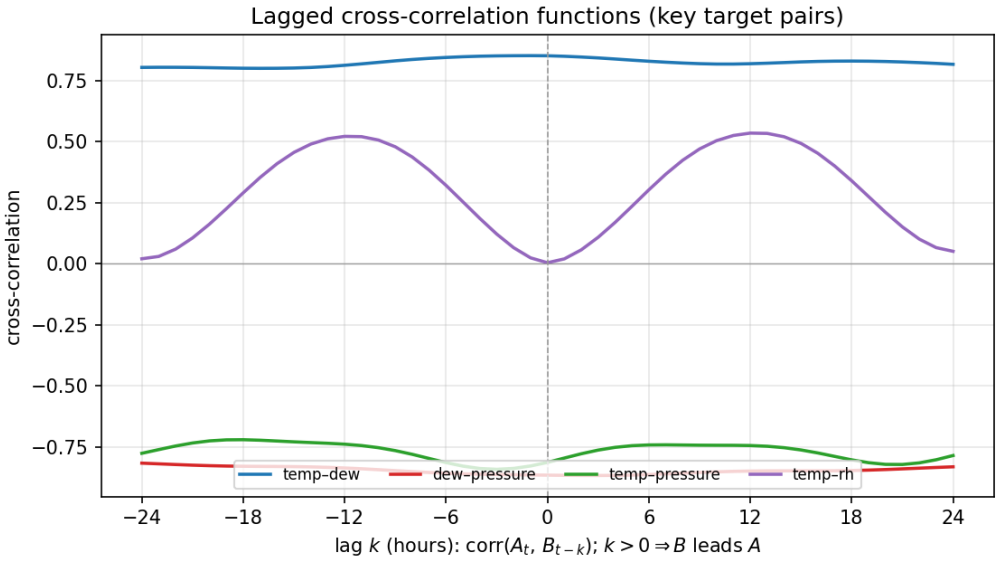


Figure 3.12: Cross-correlation functions for the key target pairs over lags $k = -24 \ldots 24$ h, where $r(k) = \mathrm{corr}(A_t, B_{t-k})$ and $k > 0$ means B leads A. Note `temp`–`rh`, flat at lag 0 but peaking near $k = 12$ h.

3.7 Key Findings from EDA

- **Complete data** — no missing values or gaps, so no imputation is needed (Chapter 4 interpolation is a safeguard only).
- **Diurnal cycle dominates** — for a 24-hour forecast the within-horizon periodicity is the daily cycle (strong in `temp`, `rh`, wind); the annual cycle acts as a slowly-varying level, and day-of-week shows *no* weekly seasonality ⇒ add cyclical hour-of-day and day-of-year encodings, but no weekday feature.
- **Strong inter-variable correlation** (temp/dew/pressure) ⇒ a multivariate model with lagged cross-features is appropriate.
- **Circular wind direction** — the wind rose shows direction is non-uniform and wraps at 360° ⇒ decompose to Cartesian `wx`/`wy` rather than feeding a raw angle.
- **Skewed/bounded targets** — `cloud` is bimodal and the wind components `wx`/`wy` take zero/near-zero values when calm ⇒ MAPE is unsafe for these, MAE is the primary metric, and `cloud` is expected to be hardest.

- **Genuine extremes** in the wind components ⇒ retained (not clipped, except the non-negativity guard on wind speed).
- **Stationary in levels** — ADF and KPSS agree the targets have no unit root (Table 3.3); the borderline KPSS cases reflect deterministic annual seasonality ⇒ no differencing ($d = 0$).
- **Low AR order + daily lag** — the PACF cuts off after a few lags with a lag-24 spike ⇒ a modest lag order augmented by daily-cycle terms suffices (confirmed by the lag-order selection in Chapter 5).
- **Lagged cross-dependence** — strong off-diagonal cross-correlations among `temp`/`dew`/`pressure` (Fig. 3.11) ⇒ a multivariate model can exploit structure unavailable to univariate ones.

Chapter 4

Data Preprocessing

The same preprocessing is applied to every model so the comparison is fair.

4.1 Data Cleaning

The `time` column is parsed to a timestamp, set as the index, sorted ascending, and resampled to a strict hourly frequency. As a sensor-error guard, wind speed is clipped to be non-negative,

$$\texttt{wind_speed} \leftarrow \max(\texttt{wind_speed}, 0),$$

which removes spurious sentinel values (e.g. -9999).

4.2 Missing Value Handling

Any incidental gaps are filled by time-based interpolation followed by forward- and back-filling:

$$\tilde{x}_t = \text{interp}(x)_t, \qquad \text{then } \texttt{ffill} \rightarrow \texttt{bfill}.$$

On the present dataset there are no gaps, so this step is a safeguard.

4.3 Wind Decomposition

Wind direction is circular and not directly learnable; it is converted to Cartesian components,

$$\texttt{wx} = \texttt{wind_speed} \cdot \cos\Big(\text{rad}(\texttt{wind_dir})\Big), \tag{4.1}$$

$$\texttt{wy} = \texttt{wind_speed} \cdot \sin\Big(\text{rad}(\texttt{wind_dir})\Big), \tag{4.2}$$

so that magnitude and direction are encoded smoothly.

4.4 Cyclical Time Encoding

Hour-of-day and day-of-year are encoded with sine/cosine pairs to respect their periodicity:

$$\text{hour}_{\sin} = \sin\left(\frac{2\pi\, h}{24}\right), \qquad \text{hour}_{\cos} = \cos\left(\frac{2\pi\, h}{24}\right), \tag{4.3}$$

$$\text{doy}_{\sin} = \sin\left(\frac{2\pi\, d}{365.25}\right), \qquad \text{doy}_{\cos} = \cos\left(\frac{2\pi\, d}{365.25}\right). \tag{4.4}$$

4.5 Feature Scaling

A `StandardScaler` is fit on the *training partition only* and applied to validation and test:

$$z = \frac{x - \mu_{\text{train}}}{\sigma_{\text{train}}}.$$

All forecasts are inverted to the real scale before any metric is computed. Tree models are scale-invariant; scaling is retained for protocol consistency with the deep-learning models.

4.6 Train/Validation/Test Split

The split is strictly chronological and never shuffled (Table 4.1).

Table 4.1: Chronological data split.

Split	Years	Hours	Role
Train	2021–2023	26,280	Fit model parameters
Val	2024	8,784	Tuning / early stopping
Test	2025	8,760	Final evaluation (untouched)

4.7 Sliding-Window Generation

For machine-/deep-learning models, the series is converted to supervised windows: an input lookback of $L = 120$ hours predicts the next $H = 24$ hours,

$$\underbrace{\mathbf{X} \in \mathbb{R}^{L \times d_{\text{in}}}}_{\text{input window}} \longrightarrow \underbrace{\mathbf{Y} \in \mathbb{R}^{H \times 7}}_{\text{forecast block}}.$$

Forecasts only ever condition on observations strictly before the target window, preventing leakage.

4.8 Feature Engineering Overview

Tree models cannot exploit a raw flattened window effectively, so each $(120, d_{\text{in}})$ window is summarised into a compact feature vector (recent lags and rolling statistics per channel). Every engineered feature is a function of the 120-hour history up to the origin, preserving the same no-leakage horizon as the raw window. The full construction is detailed in Section 5.5.3.

Chapter 5

Forecasting Models

This chapter presents the five forecasting models. Each follows the same skeleton — Overview, Mathematical Formulation, and Experimental Results — is scored by the metrics defined in Section 6.2, and is validated by the post-estimation diagnostics defined next; the side-by-side comparison is collected in Chapter 6.

5.1 Post-estimation Diagnostic Tests

After a model is fitted, but before its forecasts are trusted, a battery of *post-estimation diagnostics* verifies that the model is well specified. Each test targets a single assumption, and for the white-noise and normality tests a p-value above 0.05 is the desirable outcome (the benign null is not rejected). A model only needs the diagnostics that match the assumptions it actually makes: the statistical models (VAR, SARIMA) rest on stationarity and stability and so report the full battery, whereas the machine- and deep-learning models (XGBoost, DLinear, LSTM) make no such assumptions and instead establish adequacy through held-out validation, reporting the residual-based checks where informative. Table 5.1 summarises the tests and their applicability; the backgrounds and formulas follow.

Table 5.1: Post-estimation diagnostic tests, their hypotheses, and applicability.

Test	**Null H_0**	**Property checked**	**"Good"**	**Applies to**
ADF	unit root	stationarity (pre-fit)	reject, $p \leq 0.05$	VAR, SARIMA
KPSS	stationarity	stationarity (mirror)	accept, $p > 0.05$	VAR, SARIMA
Stability	—	non-divergence of forecasts	radius < 1	VAR, SARIMA
Ljung–Box	no autocorrelation	white-noise residuals	$p > 0.05$	all models
Jarque–Bera	normal residuals	residual normality	$p > 0.05$	VAR, SARIMA; ML/DL descriptive

Augmented Dickey–Fuller (ADF). The ADF test detects a *unit root* (a stochastic trend); a unit-root series is non-stationary and must be differenced before autoregressive modelling [4]. It estimates the regression

$$\Delta y_t = \alpha + \beta t + \gamma\, y_{t-1} + \sum_{i=1}^{q} \delta_i\, \Delta y_{t-i} + \varepsilon_t, \tag{5.1}$$

and tests $H_0 \colon \gamma = 0$ (unit root) through the statistic $\mathrm{ADF} = \hat{\gamma}/\mathrm{SE}(\hat{\gamma})$, compared against Dickey–Fuller critical values. Rejecting H_0 ($p \leq 0.05$) implies stationarity. *Used by* VAR and SARIMA to fix the differencing order d; on this dataset all seven targets reject the unit root, giving $d = 0$.

KPSS. The KPSS test is the complement of ADF, with the *opposite* null hypothesis — stationarity — so running the two together separates genuine stationarity from borderline cases [7]. With $S_t = \sum_{i=1}^{t} \hat{e}_i$ the partial sums of the residuals from regressing y_t on a constant (and optional trend),

$$\mathrm{KPSS} = \frac{1}{T^2} \sum_{t=1}^{T} \frac{S_t^2}{\hat{\sigma}_\infty^2}, \tag{5.2}$$

where $\hat{\sigma}_\infty^2$ is a long-run variance estimate. A large statistic (small p) rejects stationarity. *Used by* VAR and SARIMA as a confirmatory cross-check; here KPSS marginally rejects only for `temp` and `wx`, flagging deterministic annual seasonality rather than a stochastic

trend (Section 3.6).

Stability (characteristic roots). A fitted VAR(p) — or the autoregressive part of SARIMA — is *stable*, so its forecasts stay bounded, if and only if every root z of the characteristic polynomial lies outside the unit circle [9]:

$$\det\left(\mathbf{I}_k - \mathbf{A}_1 z - \cdots - \mathbf{A}_p z^p\right) \neq 0 \quad \text{for } |z| \leq 1, \tag{5.3}$$

equivalently the companion matrix has spectral radius below one. This is a property check, not a hypothesis test (no p-value). *Used by* VAR and SARIMA; the ML/DL analogue is confirming that training converges without divergence.

Ljung–Box Q-test. This test checks whether the residuals resemble white noise by testing that their first m autocorrelations are *jointly* zero [1]; residual autocorrelation that remains signals that the lag order or model capacity is too low. With $\hat{\rho}_k$ the residual autocorrelation at lag k,

$$Q = T(T+2)\sum_{k=1}^{m} \frac{\hat{\rho}_k^2}{T-k} \;\sim\; \chi^2_{m-r} \quad (H_0 : \text{no autocorrelation}), \tag{5.4}$$

with r the number of fitted AR/MA parameters; $p > 0.05$ indicates white noise. *Used by* every model on its residuals (here at lag $m = 24$).

Jarque–Bera. This test assesses residual normality from the sample skewness S and kurtosis K; it matters because *analytical* (Gaussian) prediction intervals assume normal residuals. The statistic is

$$\mathrm{JB} = \frac{T}{6}\left(S^2 + \tfrac{1}{4}(K-3)^2\right) \;\sim\; \chi^2_2 \quad (H_0 : \text{normal}). \tag{5.5}$$

Non-normality does not bias the point forecasts but makes Gaussian intervals only approximate, which motivates the distribution-free *bootstrap* intervals reported alongside them. *Used by* VAR and SARIMA for interval validity; the ML/DL models report it (or an equivalent residual QQ-plot) descriptively.

5.2 DLinear

5.2.1 Overview

DLinear [11] is a decomposition-based linear model that challenges the prevailing trend of applying complex Transformer architectures to time series forecasting. The core insight

is that a time series can be separated into a slow-moving trend and a fast-moving seasonal component, each forecast independently with a single linear layer, then recombined. Despite extreme simplicity — only 5,892 parameters for the shared-weight variant — DLinear matches or exceeds state-of-the-art Transformers on standard benchmarks while training orders of magnitude faster.

Why we include it. DLinear is our *minimal learned baseline*: if a single linear layer per component can rival far heavier models, it sets the bar the nonlinear methods must clear. Its explicit trend/seasonal split also mirrors the structure the EDA found in every target — a smooth annual trend plus a strong diurnal cycle (Section 3.6) — making it a natural fit for this dataset.

In this project, DLinear is applied as a **multivariate multi-step** forecaster: a single model consumes the full 120×11 input window (7 weather channels + 4 cyclical time encodings) and outputs a 24×7 forecast block covering all seven target variables simultaneously. A final linear projection layer maps the 11 input channels to the 7 output targets, allowing the model to exploit cross-variable dependencies learned from data.

Two variants are evaluated:

- **DLinear-S** (Shared): one pair of trend/seasonal linear layers shared across all 11 input channels — 5,892 parameters.
- **DLinear-I** (Individual): per-channel trend/seasonal linear layers — 63,972 parameters.

5.2.2 Mathematical Formulation

Let $\mathbf{X} \in \mathbb{R}^{L \times C_{\text{in}}}$ be an input window of $L = 120$ hours and $C_{\text{in}} = 11$ features (7 weather variables + 4 cyclical time encodings). The model produces $\hat{\mathbf{Y}} \in \mathbb{R}^{H \times C_{\text{out}}}$ with $H = 24$ hours and $C_{\text{out}} = 7$ target variables.

Series Decomposition

A 1D average pooling with kernel size $k = 25$ and stride 1 extracts the trend while preserving sequence length. For a sequence of length L, the first and last time steps are replicated $p = (k-1)/2 = 12$ times and concatenated to both ends before pooling — a technique called **replication padding** used by the official implementation [11]:

$$\mathbf{X}_{\text{padded}} = [\underbrace{\mathbf{X}_0, \ldots, \mathbf{X}_0}_{p \text{ times}}, \mathbf{X}_0, \ldots, \mathbf{X}_{L-1}, \underbrace{\mathbf{X}_{L-1}, \ldots, \mathbf{X}_{L-1}}_{p \text{ times}}] \quad \in \mathbb{R}^{(L+2p) \times C_{\text{in}}} \tag{5.6}$$

$$\mathbf{T} = \text{AvgPool1d}\Big(\mathbf{X}_{\text{padded}},\ \text{kernel} = k,\ \text{padding} = 0\Big) \quad \in \mathbb{R}^{L \times C_{\text{in}}} \tag{5.7}$$

$$\mathbf{S} = \mathbf{X} - \mathbf{T} \quad \in \mathbb{R}^{L \times C_{\text{in}}} \tag{5.8}$$

At each position t, the trend is the unweighted mean of the k values in the local window:

$$\mathbf{T}_t = \frac{1}{k} \sum_{j=t-p}^{t+p} \mathbf{X}_{\text{padded},j} \tag{5.9}$$

The kernel size $k = 25$ spans just over one full diurnal cycle (24 h), causing the day–night oscillation to approximately cancel when averaged, leaving only the multi-day trend. The seasonal component $\mathbf{S}$ captures all remaining variation: diurnal cycles, short-term fluctuations, and noise.

Why replication padding. At the boundaries ($t < p$ and $t \geq L - p$), the kernel extends beyond available data. Repeating the boundary values — rather than padding with zeros — encodes a **persistence** assumption: if nothing is known beyond the edge, the most conservative guess is that conditions remain unchanged. Concretely, at the leftmost position:

$$\mathbf{T}_0 = \frac{\overbrace{\mathbf{X}_0 + \cdots + \mathbf{X}_0}^{p+1 \text{ times}} + \mathbf{X}_1 + \cdots + \mathbf{X}_p}{k} = \frac{(p+1)\,\mathbf{X}_0 + \sum_{j=1}^{p} \mathbf{X}_j}{k} \tag{5.10}$$

All $k = 25$ positions in the kernel contain real observations or faithful copies; no position is underweighted. The first and last 12 positions (24 of 120 h, 20% of the lookback window) are affected; the rightmost 12 are the hours closest to the forecast origin, where the signal is most informative for short-horizon prediction.

Dual Linear Projection

Each component is independently mapped from the 120 h lookback to the 24 h forecast horizon via a single linear layer:

$$\hat{\mathbf{Y}}_t = \mathbf{W}_t\,\mathbf{T} + \mathbf{b}_t \quad \in \mathbb{R}^{H \times C_{\text{in}}} \qquad \text{(Trend forecast)} \tag{5.11}$$

$$\hat{\mathbf{Y}}_s = \mathbf{W}_s\,\mathbf{S} + \mathbf{b}_s \quad \in \mathbb{R}^{H \times C_{\text{in}}} \qquad \text{(Seasonal forecast)} \tag{5.12}$$

where $\mathbf{W}_t, \mathbf{W}_s \in \mathbb{R}^{H \times L}$ and $\mathbf{b}_t, \mathbf{b}_s \in \mathbb{R}^{H}$.

- **DLinear-S**: a single pair $(\mathbf{W}_t, \mathbf{W}_s)$ is shared across all C_{in} channels.
- **DLinear-I**: each channel $c \in \{1, \ldots, C_{\text{in}}\}$ has its own pair $\mathbf{W}_t^{(c)}, \mathbf{W}_s^{(c)} \in \mathbb{R}^{H \times L}$, implemented as C_{in} separate `nn.Linear(L, H)` modules.

Combine and Project

$$\hat{\mathbf{Y}}_{\text{raw}} = \hat{\mathbf{Y}}_t + \hat{\mathbf{Y}}_s \quad \in \mathbb{R}^{H \times C_{\text{in}}} \tag{5.13}$$

$$\hat{\mathbf{Y}} = \hat{\mathbf{Y}}_{\text{raw}}\, \mathbf{W}_{\text{proj}} \quad \in \mathbb{R}^{H \times C_{\text{out}}} \tag{5.14}$$

$\mathbf{W}_{\text{proj}} \in \mathbb{R}^{C_{\text{in}} \times C_{\text{out}}}$ maps the C_{in} input channels to the C_{out} target variables. This projection is linear and data-driven: the model learns weighted combinations of input channels for each target variable (e.g. wx and wy may contribute to predicting each other, temperature helps predict dew point).

Parameter count:

- DLinear-S: $2 \times (LH + H) \times 1 + C_{\text{in}}C_{\text{out}} = \mathbf{5{,}892}$
- DLinear-I: $2 \times (LH + H) \times C_{\text{in}} + C_{\text{in}}C_{\text{out}} = \mathbf{63{,}972}$

Why Decomposition Helps

The decomposition acts as an **inductive bias**. The trend branch only needs to learn a smooth continuation ("is it getting warmer or cooler over the next 24 hours?"), while the seasonal branch only needs to capture the repeating diurnal pattern ("at what hour does the daily peak occur?"). By separating these two structurally different tasks, each linear layer solves a simpler problem than one layer would face on the raw mixed signal.

Concrete example (first 48 h of 2021, temperature):

Table 5.2: Decomposition of the first 48 hours of temperature (2021).

Component	Value range	Meaning
Raw temp	$9.4 \rightarrow 18.1\,°\text{C}$	Day–night oscillation on warming trend
Trend ($k = 25$)	$5.4 \rightarrow 11.4\,°\text{C}$	Smooth multi-day warming of $\sim$6 °C
Seasonal	$+4.0$ at noon, -3.0 at midnight	Clean 24 h cycle, $\sim \pm 5\,°\text{C}$ amplitude

5.2.3 Architecture and Forecasting Pipeline

Figure 5.1 depicts the DLinear architecture — the dataflow inside the model — and Figure 5.2 the end-to-end workflow from raw CSV to final evaluation.

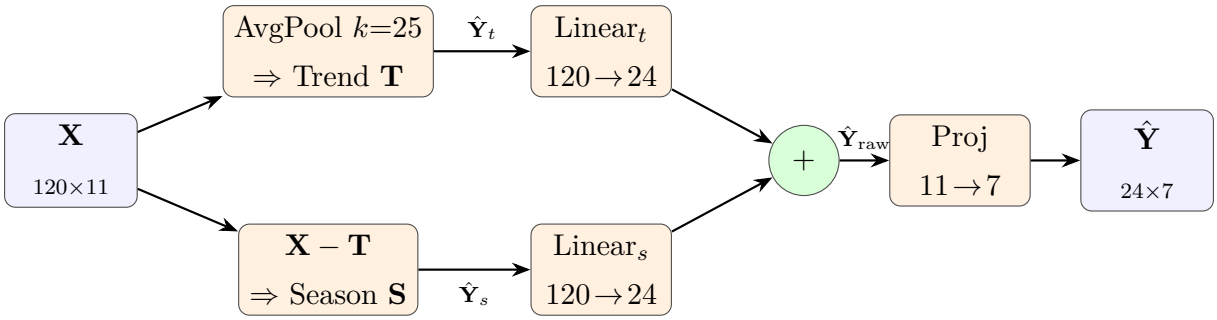


Figure 5.1: DLinear architecture. The 120 × 11 input window is split by a moving-average kernel (k=25, replication padding) into a trend $\mathbf{T}$ and a seasonal residual $\mathbf{S}$; each is mapped to the 24-step horizon by its own linear layer, the two forecasts are summed, and a learned cross-channel projection $\mathbf{W}_{\text{proj}} \in \mathbb{R}^{11\times 7}$ produces the final 24 × 7 forecast. DLinear-S shares $(\mathbf{W}_t, \mathbf{W}_s)$ across channels; DLinear-I gives each channel its own pair.

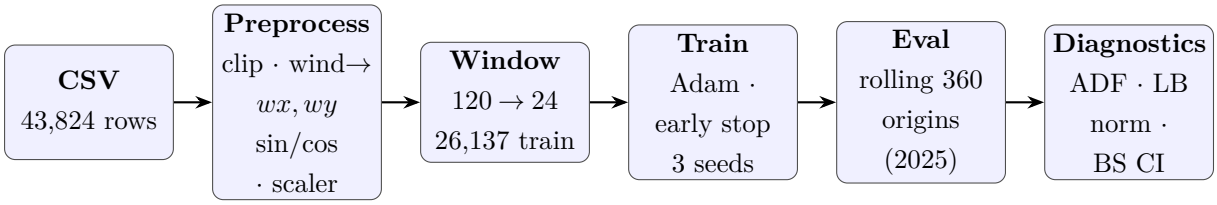


Figure 5.2: DLinear end-to-end workflow: from raw hourly data to the rolling-origin forecast and its post-estimation diagnostics.

The pipeline comprises five stages:

1. **Preprocessing** — parse time index, clip negative wind speeds (sensor-error guard), decompose wind direction to Cartesian components (wx, wy), interpolate any gaps, add 4 cyclical time encodings (sin/cos of hour-of-day and day-of-year), apply StandardScaler fitted on train only.

2. **Window generation** — slide a 120 h lookback window across the series at stride 1 to produce (X, Y) pairs of shape $(120, 11) \to (24, 7)$. Chronological split: 2021–2023 train, 2024 val, 2025 test.

3. **Training** — feed windows through the DLinear model; optimize MSE with Adam and ReduceLROnPlateau; early-stop on validation loss. Three seeds trained per variant.

4. **Evaluation** — rolling-origin over 2025: advance origin in 24 h non-overlapping steps (360 origins), invert StandardScaler, compute MAE/RMSE overall, per-variable, and per-horizon.

5. **Diagnostics** — the shared post-estimation checks (Section 5.1): ADF stationarity, Ljung–Box residual autocorrelation, residual normality, empirical forecast intervals, bootstrap CIs, and training stability.

5.2.4 Training Procedure

One model, all targets. Unlike XGBoost (which trains one regressor per target), a single DLinear model predicts all seven targets jointly. The final projection layer $\mathbf{W}_{\text{proj}}$ learns cross-target relationships directly from the training data.

Data usage. The model trains on the full 2021–2023 sliding windows at stride 1 (26,137 windows). Every window conditions strictly on observations before the target horizon, preventing leakage. No subsampling or stride reduction is applied — the model is fast enough to use all available windows.

Validation and early stopping. Training minimizes MSE with the Adam optimizer [8]. A ReduceLROnPlateau scheduler halves the learning rate when validation loss fails to improve for 5 epochs. Early stopping terminates training when validation loss does not improve for 10 consecutive epochs, capped at 50 epochs total. All hyperparameters (Table 5.3) are fixed a priori — there is no grid search — and the 2025 test set is never seen during training.

Convergence. On an NVIDIA GeForce RTX 4060 GPU, DLinear-S converges in ∼85 s per seed to a mean validation MSE of $0.3538 \pm 1.6 \times 10^{-4}$ (3 seeds). DLinear-I converges in ∼233 s per seed to $0.3451 \pm 5.7 \times 10^{-4}$. Both plateau smoothly without divergence: validation loss slopes over the last 10 epochs are -1.5×10^{-4} (S) and -3.6×10^{-4} (I) — both effectively flat. The train–validation loss gap is small (S: -0.030; I: -0.036), indicating no overfitting.

Multi-seed training. Three seeds $\{42, 123, 456\}$ are trained and checkpoints saved for each seed per variant. The purely linear optimization landscape is near-convex, leading to highly consistent convergence regardless of random initialization: across 3 seeds, MAE standard deviation is 0.004 for S and 0.023 for I.

Table 5.3: DLinear training hyperparameters.

Parameter	**Value**
Optimizer	Adam
Learning rate	10^{-3}
Weight decay	10^{-5}
Loss	MSE
Batch size	64
LR scheduler	ReduceLROnPlateau (factor 0.5, patience 5)
Early stopping	Patience 10 epochs, max 50
Kernel size	25
Seeds	$\{42, 123, 456\}$

5.2.5 Algorithm

Algorithm 1 DLinear training and rolling-origin evaluation

Require: Hourly series $S[0..T]$; $L = 120$; $H = 24$; $V = 7$; $k = 25$; seeds $\{42, 123, 456\}$

Preprocess:
  Parse time; clip wind ≥ 0
  $wx = \text{speed} \cdot \cos(\text{dir})$, $wy = \text{speed} \cdot \sin(\text{dir})$
  Add sin/cos hour-of-day, day-of-year
  Chronological split: train(2021–23), val(2024), test(2025)
  Fit StandardScaler on train, transform all
Generate sliding windows (stride 1):
  $X[i] = S[i : i + L] \in \mathbb{R}^{L \times 11}$, $Y[i] = S[i + L : i + L + H] \in \mathbb{R}^{H \times 7}$
for each seed $s \in \{42, 123, 456\}$ **do**
  Init. DLinear($k = 25$, individual)
  for epoch = 1 to 50 **do**
    $\mathbf{T} = \text{AvgPool1d}(\mathbf{X})$; $\mathbf{S} = \mathbf{X} - \mathbf{T}$
    $\hat{\mathbf{Y}} = (\mathbf{W}_t\mathbf{T} + \mathbf{b}_t) + (\mathbf{W}_s\mathbf{S} + \mathbf{b}_s)$
    $\hat{\mathbf{Y}} \leftarrow \hat{\mathbf{Y}}\,\mathbf{W}_{\text{proj}}$
    $\mathcal{L} = \text{MSE}(\hat{\mathbf{Y}}, \mathbf{Y})$
    Adam step; update $\mathbf{W}_t, \mathbf{W}_s, \mathbf{W}_{\text{proj}}$
    if val loss stagnant 5 epochs **then**
      $\eta \leftarrow \eta \times 0.5$
    end if
    if val loss stagnant 10 epochs **then**
      break
    end if
  end for
  Save best checkpoint
end for
Rolling-origin evaluation (test 2025, step 24 h, $N \approx 360$):
for each origin t **do**
  $\hat{\mathbf{Y}}_t = \text{model}(S[t - L : t]) \in \mathbb{R}^{24 \times 7}$
  Inverse-transform $\hat{\mathbf{Y}}_t$ to real scale
end for
Aggregate: overall MAE/RMSE; per-variable; per-horizon; bootstrap 95% CI

5.2.6 Post-estimation Diagnostics

Assumptions and their validity. Being a linear, learned model, DLinear drops the parametric assumptions of the statistical models — it requires no stationarity, no unit-root condition, and no characteristic-root stability. Its working assumptions are structural, and we note how each holds:

- **Additive trend + seasonal decomposition is meaningful.** *Satisfied* — the EDA shows exactly a smooth trend plus a strong diurnal cycle (Section 3.6), which is what the moving-average split separates.
- **A linear map from the 120-h window to the 24-h horizon is adequate.**

Partially — this is DLinear's main limitation: it caps accuracy on the nonlinear targets (`cloud`, gusts) while staying adequate for the smooth ones. It is not "fixed" but accepted as the model's design trade-off, and quantified by the per-variable errors.

- **Prediction bands need no Gaussian residuals.** *Handled* — the residuals are non-normal (below), so empirical residual quantiles are used for the bands instead of Gaussian ones.

The pre-fit ADF and stability tests of Section 5.1 therefore do not apply; adequacy is instead established empirically through convergence, residual whiteness (Ljung–Box), and residual normality on the 2025 test residuals.

Table 5.4: DLinear post-estimation diagnostics (2025 test).

Check	**DLinear-S**	**DLinear-I**	**Interpretation**
Convergence	PASS (slope -1.5×10^{-4})	PASS (slope -3.6×10^{-4})	Flat val loss, fully converged
Ljung–Box (lag 24)	76/168 (45%)	90/168 (54%)	Autocorrelation at short horizons (h = 1..6); pressure & dew_point worst
Normality	FAIL ($p \approx 0$)	FAIL ($p \approx 0$)	Expected — cloud bimodal, wind heavy-tailed
Forecast int. (95%)	Temp ±1.85 → ±2.90 °C	Temp ±1.64 → ±2.76 °C	Monotonic widening; pressure widens ×5.3 in I
Bootstrap CI (MAE)	±3.5%	±3.7%	Narrow (< 5%), reliable metrics

Ljung–Box: Residual autocorrelation is present in roughly half of the 168 horizon-variable pairs, concentrated at short horizons ($h = 1..6$) and in pressure and dew_point — variables the EDA identifies as having long memory (Chapter 3). DLinear-I's individual weights partially mitigate this (54% pass vs. 45% for S).

Residual normality: Universally rejected, as expected. Cloud residuals are U-shaped (bimodal target), wind residuals are heavy-tailed (gust events). Gaussian-based confidence intervals would be unreliable; empirical quantile-based intervals are used instead.

Forecast intervals: All variables show monotonic widening from $h = 1$ to $h = 24$. Pressure widens most aggressively (×5.3 in DLinear-I) due to its long-memory structure — the model's uncertainty about pressure grows faster than for temperature (×1.6).

Bootstrap CI: MAE confidence interval width under ±4% for both variants confirms stable metric estimates.

5.2.7 Experimental Results

Evaluated by rolling-origin over the 2025 test year ($N = 360$ non-overlapping origins, step = 24 h, real scale). Both variants substantially outperform the naive repeat-last-value baseline (MAE = 6.71). Values are ensemble-mean across 3 seeds $\{42, 123, 456\}$.

Overall Performance

Table 5.5: Overall test-set performance (2025, real scale, 3-seed ensemble).

Model	MAE	RMSE	MAE 95% CI	RMSE 95% CI	Params	Train
DLinear-S	5.79	12.27	[5.59, 5.99]	[11.73, 12.79]	5,892	85 s
DLinear-I	5.64	12.19	[5.41, 5.82]	[11.63, 12.70]	63,972	233 s

DLinear-I improves over DLinear-S by +2.7% (MAE) and +0.6% (RMSE), at the cost of 10.9× more parameters. This aligns with the original paper's findings [11]: individual weights remove the forced compromise of sharing one temporal pattern across physically diverse channels, allowing each variable to learn its own dynamics. The RMSE/MAE ratio is ∼2.1 for both variants, indicating occasional large errors driven by cloud cover and wind extremes. Bootstrap 95% CIs (1000 resamples) are narrow (width $< 4\%$), confirming stable metric estimates.

Per-Variable Error

Table 5.6: Per-variable MAE (real scale, 2025).

Target	S MAE	I MAE	Unit	I vs. S
temperature	1.22	1.18	°C	+2.7%
dew_point	1.28	1.19	°C	+6.4%
surface_pressure	1.33	1.13	hPa	+15.2%
relative_humidity	5.70	5.64	%	+1.1%
wx	4.47	4.39	km/h	+1.8%
wy	3.10	3.07	km/h	+1.1%
cloud_cover	23.41	23.00	%	+1.8%

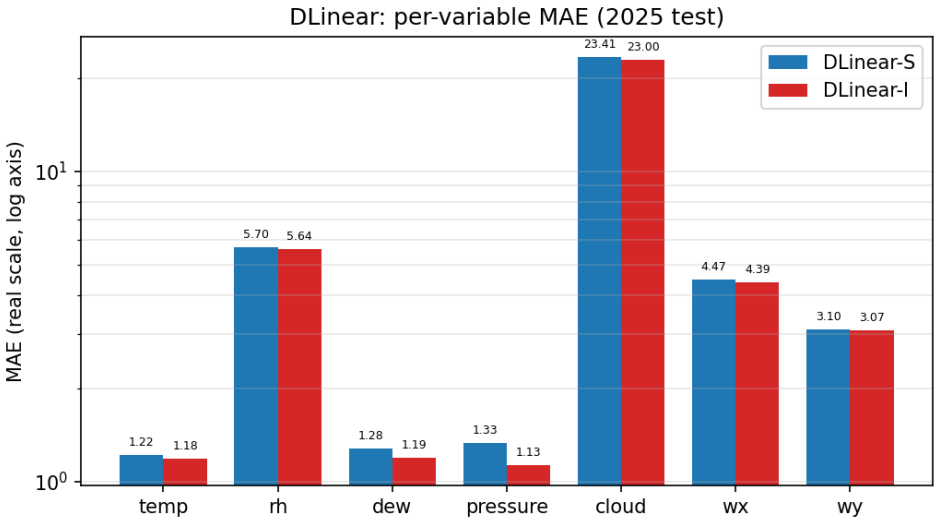


Figure 5.3: DLinear: per-variable MAE for both variants (2025 test, log axis). Cloud cover dominates the error budget; DLinear-I edges out DLinear-S on every target.

DLinear-I improves on every variable (Fig. 5.3). The largest gain is surface pressure (+15.2%), whose slow-varying, weak-diurnal-cycle dynamics differ fundamentally from temperature and humidity (strong 24 h cycle). Shared weights force a compromise across these diverse patterns; individual weights remove this constraint, confirming the paper's channel-independence hypothesis.

Cloud cover remains the hardest target across all models (MAE ∼23%), driven by its bimodal distribution (clustered at 0% and 100%) and weak linear correlation with other variables.

Per-Horizon Error Growth

Error grows approximately 2.0× from $h = 1$ to $h = 24$ (DLinear-S):

Table 5.7: Per-horizon MAE (DLinear-S, 2025 test).

Horizon	$h = 1$	$h = 4$	$h = 8$	$h = 12$	$h = 16$	$h = 20$	$h = 24$
MAE	3.46	4.58	5.07	5.77	6.75	6.82	7.03

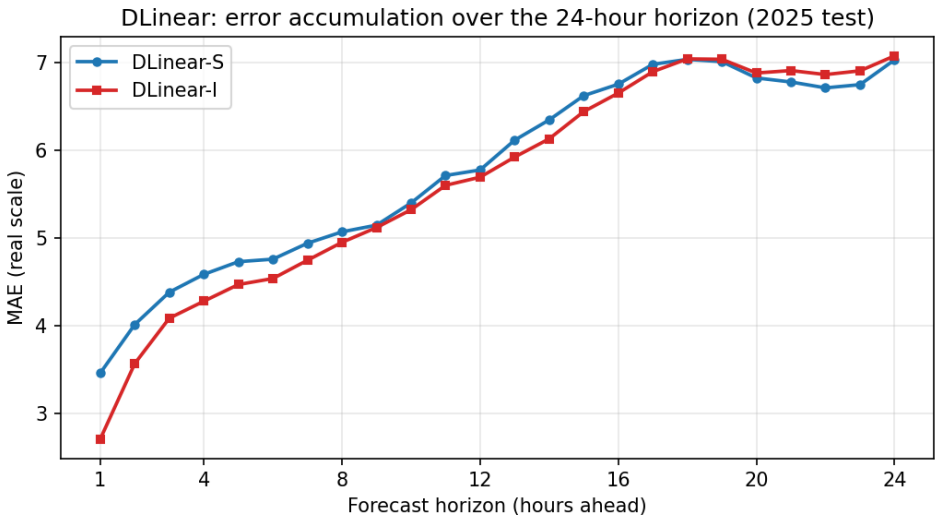


Figure 5.4: DLinear: error accumulation over the 24-hour horizon (2025 test), both variants. Error roughly doubles from h=1 to h=24 and the two variants track each other closely.

The error growth (Fig. 5.4) is approximately linear with horizon. The plateau between $h = 20$ and $h = 24$ suggests the model's uncertainty saturates beyond roughly 20 hours — additional lookback or auxiliary features may be needed for long-horizon forecasts.

5.3 SARIMA

5.3.1 Overview

SARIMA (Seasonal ARIMA) models a *univariate* series with autoregressive (AR), integrating/differencing (I), and moving-average (MA) operators acting at both ordinary and seasonal lags [1, 7]. It is the seasonal member of the broader **ARIMA family** — AR, MA, ARMA, ARIMA, SARMA, SARIMA — whose members are recovered by switching individual orders on or off.

In this project the 24-hour multivariate forecast is produced by **seven independent univariate models**, one per target (`temp,rh,dew,pressure,cloud,wx,wy`); each forecasts its own variable for the next 24 hours and the seven outputs are stacked into the required 24×7 block. Unlike VAR it does not directly model cross-variable dependencies such as the temperature–dew-point–humidity–pressure coupling. The distinguishing choice here is that we do **not** impose one fixed order on all seven series. Each target searches *its own* candidate grid — drawn from across the ARIMA family and bounded by that variable's own ACF/PACF signature (Section 3.6) — and keeps whichever family member minimises rolling-origin validation MAE (Section 5.3.3). A weakly periodic target is free to select a non-seasonal model; a persistent, strongly diurnal one can select a seasonal one.

Why we include it. SARIMA is the *classical univariate reference*: the textbook approach to a single seasonal series, giving a transparent, assumption-driven baseline against which the multivariate (VAR) and nonlinear (XGBoost, LSTM) models are judged. Per-feature selection makes it the *strongest fair* univariate baseline — each series receives the family member its own autocorrelation justifies rather than a one-size-fits-all order — so the remaining gap to the multivariate models cleanly isolates the value of modelling cross-variable structure, not an artefact of an under-tuned baseline.

5.3.2 Mathematical Formulation

A SARIMA$(p, d, q)(P, D, Q)_s$ model has non-seasonal orders (p, d, q), seasonal orders (P, D, Q), and seasonal period s. With B the backshift operator ($B^k y_t = y_{t-k}$) and ε_t white noise,

$$\Phi_P(B^s)\,\phi_p(B)\,(1-B)^d(1-B^s)^D y_t = c + \Theta_Q(B^s)\,\theta_q(B)\,\varepsilon_t, \tag{5.15}$$

where ϕ_p, θ_q are the non-seasonal AR/MA polynomials, Φ_P, Θ_Q their seasonal counterparts, and c a constant. Every member of the ARIMA family is a special case of Eq. (5.15) (Table 5.8); the search in Section 5.3.3 lets each target pick the member its data support. For hourly weather the seasonal period is the daily cycle, $s = 24$.

Table 5.8: ARIMA-family members as special cases of Eq. (5.15).

Member	**Restriction**	**Captures**
AR(p)	$q = 0$, no seasonal, $d = 0$	own-lag persistence
MA(q)	$p = 0$, no seasonal, $d = 0$	short shock memory
ARMA(p, q)	no seasonal, $d = 0$	persistence + shocks
ARIMA(p, d, q)	$d > 0$, no seasonal	trending / unit-root series
SARMA	$d = D = 0$, seasonal $P, Q > 0$	level-stationary + daily cycle
SARIMA	$d > 0$ or $D > 0$, seasonal	trending + seasonal

5.3.3 Model Identification and Selection

Rather than assume an order, each target's model is *identified* from its own autocorrelation structure (Box–Jenkins), then *selected* from a small per-feature grid by validation accuracy.

Step 1 — Box–Jenkins identification from ACF/PACF. The orders are read off the autocorrelation function (ACF) and partial autocorrelation function (PACF) measured on the 2021–2023 training split (Section 3.6):

- **Differencing $d = D = 0$.** ADF rejects a unit root and KPSS fails to reject stationarity for all seven targets, so the series are *level-stationary*; the annual cycle is deterministic, not a unit root, so no differencing is applied (forcing d=1 would only inject noise).
- **Non-seasonal p from the PACF cutoff** — the lag beyond which the PACF falls inside its confidence band — and **non-seasonal q from the ACF cutoff**. The targets' PACFs drop off within 1–3 lags, giving small p; a single q absorbs the remaining short-lag shock structure.
- **Seasonal P, Q from spikes at the seasonal lags** $(24, 48, \dots)$: a spike in the PACF at lag 24 argues for a seasonal AR term (P=1); a spike in the ACF at lag 24 with *no* matching PACF spike argues instead for a seasonal MA term (Q=1).
- **Seasonal period $s = 24$** — the diurnal cycle — identified by the clean daily oscillation in the time plots and the lag-24 autocorrelation spike.

Step 2 — per-feature candidate grids. Identification does not pin a single order; it bounds a *small neighbourhood* of plausible ones. Each target therefore gets its *own* four-candidate grid (Table 5.9) spanning that neighbourhood and, where the seasonal signal is weak, a fully non-seasonal alternative. This is the key departure from a one-grid-fits-all baseline: a variable searches only the orders its ACF/PACF justify, so the search is small, targeted, and free to let a non-seasonal model win where the daily cycle is faint.

Table 5.9: Per-feature candidate grids. Each row is one target's search space, bounded by its own ACF/PACF signature; seasonal period $s = 24$ throughout, and $(0, 0, 0)$ denotes a non-seasonal candidate. Validation MAE picks the winner (Table 5.10).

Target	**ACF/PACF signature (train)**	**Candidates** $(p, 0, q) \times (P, 0, Q)_{24}$
`temp`	slow ACF decay; negative PACF spike at lag 24	$(2,0,0)(1,0,0)$; $(3,0,0)(1,0,0)$; $(2,0,1)(1,0,1)$; $(3,0,1)(1,0,1)$
`rh`	moderate persistence, faster decay	$(2,0,0)(1,0,0)$; $(2,0,1)(1,0,1)$; $(2,0,0)(0,0,1)$; $(1,0,1)(1,0,1)$
`dew`	ACF barely decays; PACF ≈ 0 at lag 24	$(2,0,0)(0,0,1)$; $(3,0,0)(0,0,1)$; $(2,0,1)(1,0,1)$; $(2,0,0)(0,0,0)$
`pressure`	strongest seasonal PACF spike (-0.27)	$(2,0,0)(1,0,0)$; $(1,0,1)(1,0,0)$; $(2,0,1)(1,0,1)$; $(1,0,1)(1,0,1)$
`cloud`	weak daily cycle ($\text{acf}_{24} \approx 0.42$, $\text{pacf}_{24} \approx 0$)	$(1,0,1)(0,0,0)$; $(2,0,0)(0,0,1)$; $(2,0,1)(0,0,1)$; $(1,0,1)(0,0,1)$
`wx`	moderate persistence, real daily cycle	$(2,0,0)(1,0,0)$; $(2,0,1)(1,0,1)$; $(2,0,0)(0,0,1)$; $(1,0,1)(1,0,1)$
`wy`	weakest seasonality ($\text{acf}_{24} \approx 0.38$, $\text{pacf}_{24} \approx 0$)	$(1,0,1)(0,0,0)$; $(2,0,0)(0,0,0)$; $(2,0,0)(0,0,1)$; $(2,0,1)(0,0,1)$

Step 3 — selection by validation MAE. For each candidate we fit a `SARIMAX` on 2021–2023, score it by rolling-origin forecasting on the 2024 validation split (horizon H=24, step 24), and keep the lowest-validation-MAE candidate, using training AIC as a tie-break. The procedure runs independently for all seven targets; `model_selection.csv` records every candidate's score as the selection receipt. Table 5.10 lists the winners.

Table 5.10: Selected model per target (lowest 2024 validation MAE from each grid; $s = 24$, $d = D = 0$). All seven converged on the final 2021–2023 fit.

Target	**Selected model**	**Val. MAE**	**AIC**
`temp`	$\text{SARMA}(3,0,1)(1,0,1)_{24}$	1.37	42 928
`rh`	$\text{SARMA}(1,0,1)(1,0,1)_{24}$	5.79	133 743
`dew`	$\text{SARMA}(2,0,1)(1,0,1)_{24}$	1.13	38 800
`pressure`	$\text{SARMA}(1,0,1)(1,0,1)_{24}$	1.18	8 895
`cloud`	$\text{SARMA}(2,0,1)(0,0,1)_{24}$	23.58	228 480
`wx`	$\text{SARMA}(2,0,1)(1,0,1)_{24}$	4.56	118 438
`wy`	$\text{SARMA}(2,0,1)(0,0,1)_{24}$	3.13	114 477

What the data chose. Every target selected a seasonal ARMA model (SARMA, d=D=0), confirming the daily term earns its place across the board — but the *orders*

differ by variable, exactly as their ACF/PACF differ. The very persistent, strongly diurnal series (`temp`, `pressure`) took a seasonal AR term (P=1); `dew`, whose PACF is flat at lag 24, took a pure seasonal MA term; the noisiest, weakly periodic targets (`cloud`, `wy`) dropped the seasonal AR entirely and kept only a seasonal MA — the cheapest way to acknowledge their faint daily cycle. For `rh` the leaner $(1, 0, 1)$ non-seasonal core beat the $(2, 0, 1)$ one on validation MAE (and the latter failed to converge), so the simpler model won.

Why daily (s=24) and not annual seasonality. The EDA shows *two* periodicities — a 24-hour diurnal cycle and a 365-day annual cycle — but only the daily one is usable as a seasonal period here. Three reasons: (i) a yearly seasonal lag would be $s = 8760$ hours, computationally infeasible (the seasonal operators act at lag 8760 on a 26,280-point series); (ii) the forecast horizon is only 24 hours, so within any one window the annual cycle is effectively a constant level — it shifts the mean but does not *repeat*, carrying no within-horizon periodicity; and (iii) the slow annual drift that remains is absorbed by the constant term and by conditioning on recent history at each rolling origin. In short, s=24 captures the only cycle that recurs inside a 24-hour forecast; the annual cycle is handled as slowly-varying level, exactly as the ML/DL models handle it through their day-of-year encodings.

5.3.4 Training Procedure

One model per target. Seven independent models are trained, each using only the past values of its own series — a univariate statistical baseline rather than a multivariate learner.

Data usage. Parameters are fitted on 2021–2023; 2024 is used for model selection. During test forecasting the model conditions on true pre-test history without changing its fitted parameters. With `--fit-through-val` the selected model is refitted on 2021–2024 before forecasting 2025.

Convergence safeguards. Fitting maximises the Gaussian likelihood by `L-BFGS` (`maxiter`= 100). Stationarity/invertibility are left unconstrained by default (faster, and the level-stationary targets rarely need them), but `pressure` — convergence-fussy with free roots in earlier runs — enforces both. A filter-only (`low_memory`) fit avoids the Kalman smoother's large state arrays, keeping the long hourly series tractable. All seven selected models converged on the final fit.

Determinism. Fitting is deterministic for a fixed dataset and configuration; the random seed affects only the bootstrap metric intervals, not the model.

Table 5.11: Training and evaluation settings (per-feature ARIMA-family pipeline).

Parameter	**Value**
Implementation	`statsmodels SARIMAX`
Target strategy	one univariate model per target (7 total)
Candidate grid	per-feature, 4 each (Table 5.9)
Differencing	$d = D = 0$ (level-stationary)
Seasonal period	$s = 24$ h
Trend	constant (`c`)
Selection metric	2024 validation MAE (AIC tie-break)
Selection horizon / step	24 h / 24 h
Optimiser / iterations	`L-BFGS`, `maxiter`= 100
Stationarity / invertibility	not enforced (`pressure`: enforced)
Initialization	`approximate_diffuse`, `low_memory`
Fitting scope	2021–2023 (optional refit through 2024)
Bootstrap samples	1000 (metrics only, seed 42)

5.3.5 Forecasting Procedure

Training and prediction are separated: training selects and saves one fitted model per target; prediction reloads those models and evaluates the 2025 rolling-origin forecasts without refitting (Figure 5.5). At each origin t the pipeline forecasts $t{+}1, \ldots, t{+}24$ for every target, stacks the seven univariate forecasts into a 24×7 block, inverts to the real scale, and scores against the actuals.

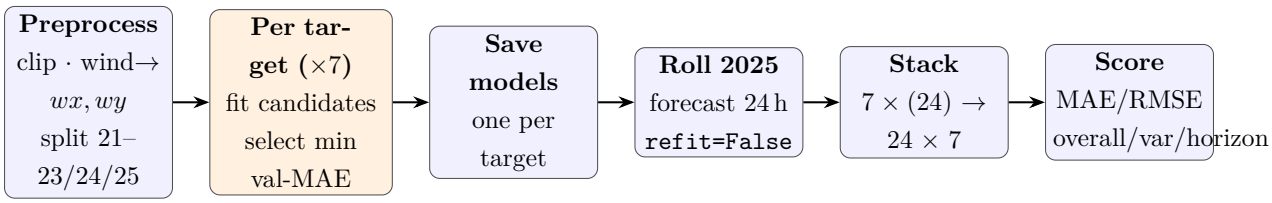


Figure 5.5: SARIMA workflow: seven univariate models are selected and saved once, then rolled over 2025 and stacked into the multivariate forecast block.

Algorithm 2 SARIMA selection and rolling-origin evaluation

Require: Hourly targets V=7; horizon H=24; step 24; candidate grid

Preprocess; chronological split train(2021–23)/val(2024)/test(2025)
for each target $v \in \{1, \ldots, 7\}$ **do**
    for each candidate $(p, d, q)(P, D, Q)_{24}$ **do**
        Fit on train; roll-forecast val; record validation MAE
    end for
    Select min-MAE candidate (AIC tie-break); refit on fit scope; save
end for
for each 2025 origin t (step 24) **do**
    Forecast $t{+}1, \ldots, t{+}H$ per target; stack to $H \times V$
    Invert to real scale; compare with actuals
end for
Aggregate MAE/RMSE overall, per-variable, per-horizon + bootstrap CI

5.3.6 Post-estimation Diagnostics

The diagnostics (defined in Section 5.1) confirm the targets are stationary on the training split and that all seven models converged, but the fitted residuals still carry autocorrelation and are non-normal — expected for compact univariate models on complex weather data.

Table 5.12: Post-estimation diagnostics for the selected models (train fit / 2025 test).

Diagnostic	**Result**	**Interpretation**
ADF stationarity	PASS (all 7)	Train series reject the unit root; $d = 0$ justified.
Convergence	7/7	All selected models converged on the final fit.
Ljung–Box (lag 24)	FAIL (all 7)	Residual autocorrelation remains after fitting.
Jarque–Bera	FAIL (all 7)	Non-normal residuals (bounded/heavy-tailed targets).
Bootstrap CI (MAE)	$[5.84, 6.26]$	Narrow relative to the 6.04 point estimate.

Residual autocorrelation. Ljung–Box rejects white noise for every target at lag 24: the univariate models leave temporal structure (daily patterns and longer-memory regimes) that a multivariate model could partly absorb. **Normality** is likewise rejected

for all targets — cloud is bounded and bimodal, wind components heavy-tailed during gusts — so analytical Gaussian intervals are interpreted with caution.

5.3.7 Experimental Results

Evaluated by rolling-origin over the 2025 test year ($N = 365$ origins, real scale), the per-feature ARIMA-family models obtain an **overall MAE of** 6.04 (95% bootstrap CI $[5.84, 6.26]$) and **overall RMSE of** 12.63 ($[12.12, 13.17]$). This is the deterministic univariate reference point of the comparison rather than the expected winner.

Table 5.13: SARIMA — overall test metrics (real scale, 2025).

Metric	**Value**	**95% bootstrap CI**
Overall MAE	6.04	[5.84, 6.26]
Overall RMSE	12.63	[12.12, 13.17]

Table 5.14: SARIMA — per-variable error (real scale, horizon-averaged, 2025).

	`temp` (°C)	`rh` (%)	`dew` (°C)	`pressure` (hPa)	`cloud` (%)	`wx` (km/h)	`wy` (km/h)
MAE	1.35	6.14	1.20	1.24	24.45	4.63	3.29
RMSE	1.86	8.24	1.85	1.88	31.33	6.06	4.54

Temperature, dew point, and pressure are forecast accurately (MAE $\sim$ 1.2–1.4), while `cloud` is the hardest target (MAE 24.45) — matching the EDA finding that cloud cover is noisy, bounded, and weakly correlated with the other variables. Error grows with lead time, from MAE 2.23 at h=1 to 7.27 at h=24 (Fig. 5.6); uniquely among the five models it is the *best* short-range forecaster at h=1 but the weakest at every longer lead, since with no cross-variable inputs it has nothing to lean on as its own seasonal memory decays.

Table 5.15: SARIMA — selected per-horizon MAE (2025 test).

Horizon	h=1	h=4	h=8	h=12	h=16	h=20	h=24
MAE	2.23	4.48	5.51	6.40	7.17	7.16	7.27

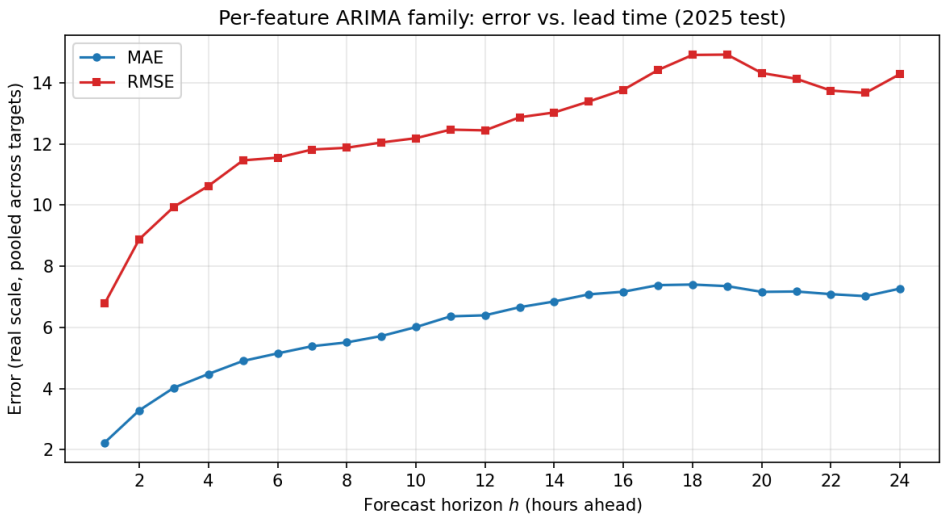


Figure 5.6: SARIMA: error accumulation over the 24-hour horizon (2025 test).

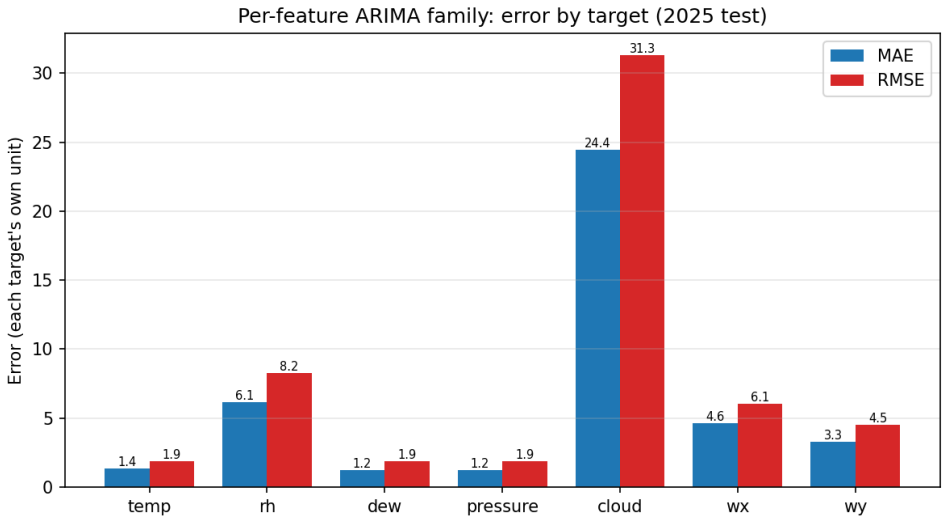


Figure 5.7: SARIMA: MAE and RMSE by target variable (2025 test).

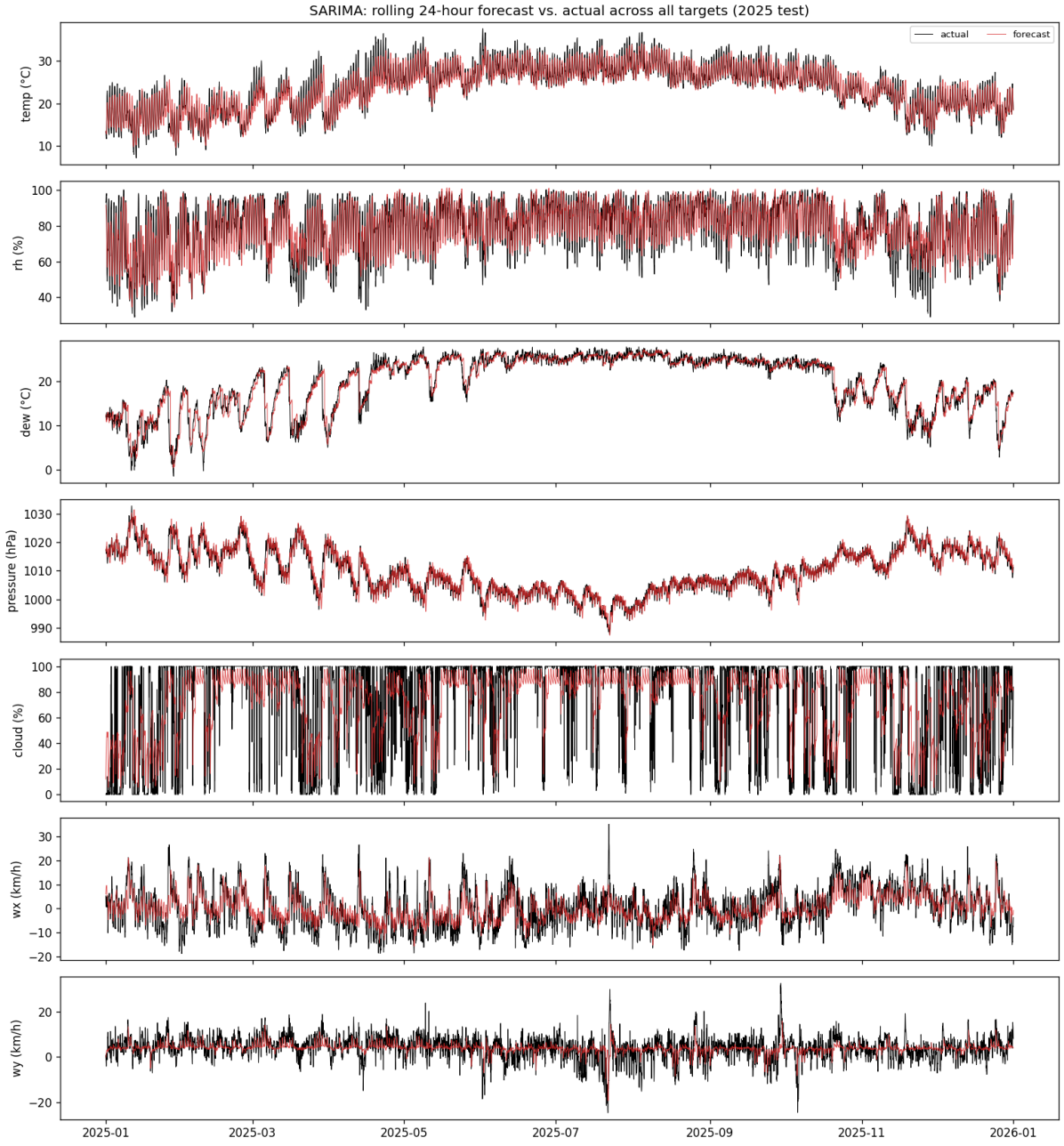


Figure 5.8: SARIMA: rolling 24-hour forecast vs. actual across all seven targets (2025 test year).

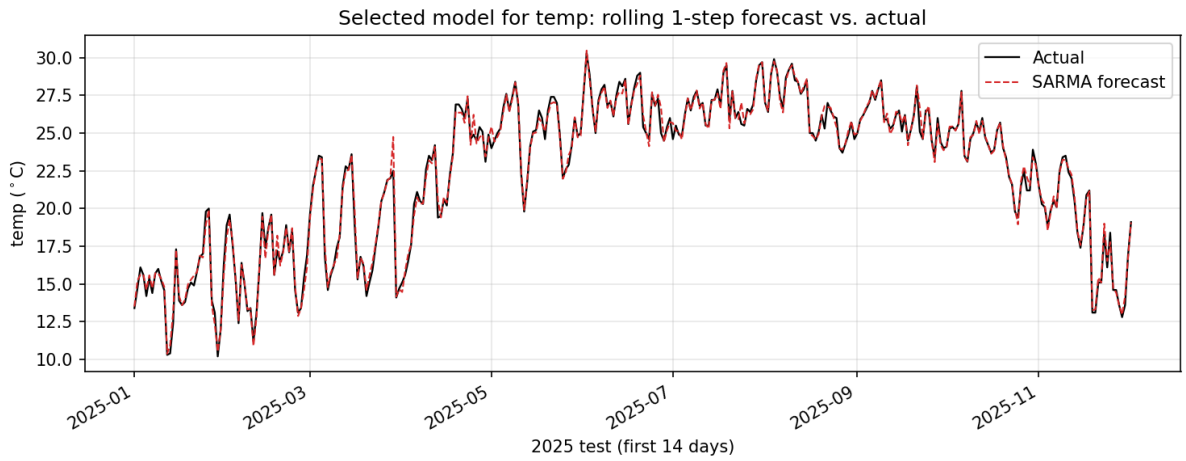


Figure 5.9: SARIMA: rolling one-step `temp` forecast vs. actual over the first two weeks of the 2025 test year (detail view).

5.4 Vector Autoregression (VAR)

5.4.1 Overview

Vector Autoregression (VAR) models the seven targets *jointly*: every variable is regressed on its own lags and the lags of all other variables, so the linear cross-dependencies seen in the EDA (e.g. the strong `temp`–`dew`–`pressure` coupling, Chapter 3) are captured directly [4, 9]. The model is multivariate, fully interpretable (each coefficient is the linear effect of one variable's lag on another), deterministic to fit (ordinary least squares), and provides analytic forecast intervals. Its limitations are that it is *linear*, its parameter count grows as k^2p, and it assumes weak stationarity — an assumption we test explicitly in Section 5.4.5.

Why we include it. VAR is chosen as the *multivariate statistical* model, the direct counterpart to univariate SARIMA. The EDA found strong lagged cross-correlations among the targets (Section 3.6) that a per-series model cannot use; VAR is the simplest model that turns those cross-dependencies into forecasts, and because the targets are level-stationary it can be fit on levels in closed form. It therefore quantifies the value of *joint* modelling at near-zero cost, and — being linear — it is the clean baseline the nonlinear learners (XGBoost, LSTM) must beat to justify their complexity.

5.4.2 Mathematical Formulation

For a k-dimensional observation $\mathbf{y}_t$ ($k = 7$) and lag order p, the VAR(p) process is

$$\mathbf{y}_t = \mathbf{c} + \sum_{i=1}^{p} \mathbf{A}_i \mathbf{y}_{t-i} + \boldsymbol{\varepsilon}_t, \qquad \boldsymbol{\varepsilon}_t \sim (\mathbf{0}, \boldsymbol{\Sigma}), \tag{5.16}$$

with intercept $\mathbf{c} \in \mathbb{R}^k$, coefficient matrices $\mathbf{A}_i \in \mathbb{R}^{k \times k}$, and white-noise innovations of covariance $\boldsymbol{\Sigma}$. Figure 5.10 depicts this predictive structure: the lagged state vectors of all seven variables are mapped through the coefficient matrices $\mathbf{A}_i$ and summed with the intercept to give the next-hour vector, which is then fed back as the newest lag to roll the forecast forward across the 24-hour horizon.

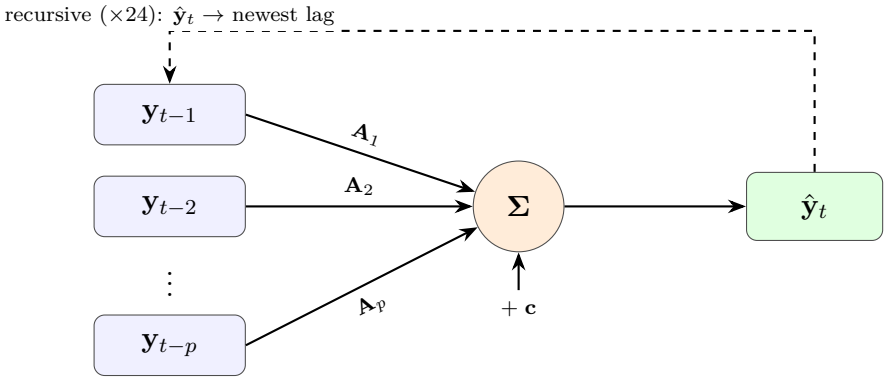


Figure 5.10: VAR prediction structure. The lagged state vectors $\mathbf{y}_{t-1}, \ldots, \mathbf{y}_{t-p}$ (each a 7-vector of `temp,rh,dew,pressure,cloud,wx,wy`) are mapped through the coefficient matrices $\mathbf{A}_1, \ldots, \mathbf{A}_p$ and summed with the intercept $\mathbf{c}$ to give the one-step forecast $\hat{\mathbf{y}}_t$; the prediction is appended as the newest lag and the window rolled forward, repeated 24 times for the full horizon.

Estimation. Because every equation shares the same regressors (the stacked lags), VAR is estimated by **equation-by-equation OLS**, which equals the Gaussian maximum-likelihood estimator. Fitting is therefore closed-form and seed-free.

Lag-order selection. The order p minimises the Akaike Information Criterion over $p \in \{1, \ldots, p_{\max}\}$,

$$\mathrm{AIC}(p) = \ln \left| \hat{\boldsymbol{\Sigma}}(p) \right| + \frac{2}{T}\, p\, k^2, \tag{5.17}$$

trading goodness of fit against the $k^2 p$ parameters.

Stability and forecast intervals. The process is stable (hence stationary) iff every root z of $\det(\mathbf{I}_k - \mathbf{A}_1 z - \cdots - \mathbf{A}_p z^p) = 0$ lies outside the unit circle. Forecasts are produced recursively from Eq. (5.16); the h-step error covariance is

$$\boldsymbol{\Sigma}(h) = \sum_{j=0}^{h-1} \boldsymbol{\Phi}_j\, \boldsymbol{\Sigma}\, \boldsymbol{\Phi}_j^{\top}, \tag{5.18}$$

with moving-average matrices $\boldsymbol{\Phi}_j$, giving the analytic band $\hat{y}_v(h) \pm z_{1-\alpha/2} \sqrt{[\boldsymbol{\Sigma}(h)]_{vv}}$. Equation (5.18) assumes Gaussian innovations — an assumption violated here (Section 5.4.5), motivating the bootstrap interval below.

5.4.3 Model Specification and Training

Stationarity and differencing order. An Augmented Dickey–Fuller (ADF) test on each training series rejects the unit-root null at $p \leq 10^{-8}$ for all seven targets (Table 5.17), so the model is fit on *levels* ($d = 0$). The borderline KPSS results for `temp` and `wx` (Chapter 3) reflect the deterministic *annual* cycle rather than a stochastic trend, which differencing would not remove.

Lag order. Minimising AIC (Eq. (5.17)) over $p_{\max} = 60$ selects $\mathbf{p = 50}$, a genuine interior minimum (the criterion rises for $p > 50$); the more parsimonious BIC and HQIC prefer $p = 26$. We adopt the AIC choice, which also yields white-noise residuals (Section 5.4.5). The model is fit once on the 2021–2023 partition (26,280 hourly rows) by OLS; being deterministic, it needs no seed ensemble (contrast XGBoost, Section 5.5.4).

Table 5.16: VAR configuration.

Setting	**Value**
Variables k	7 (`temp,rh,dew,pressure,cloud,wx,wy`)
Estimation	equation-by-equation OLS (deterministic)
Differencing d	0 (levels; ADF)
Lag search $p_{\max}$	60
Selected lag p (AIC)	50
Parameters $k(kp{+}1)$	2,457
Training rows	26,280 (2021–2023)

5.4.4 Forecasting Procedure

The end-to-end VAR pipeline — from raw hourly data through specification, estimation, and diagnostics to the rolling forecast — is summarised in Figure 5.11. At each test origin the last p observed rows seed a recursive 24-step forecast (the feedback structure of Fig. 5.10), which is inverted to the real scale before scoring; the rolling-origin loop matches the shared protocol (Section 6.1). Algorithm 3 gives the procedure in detail.

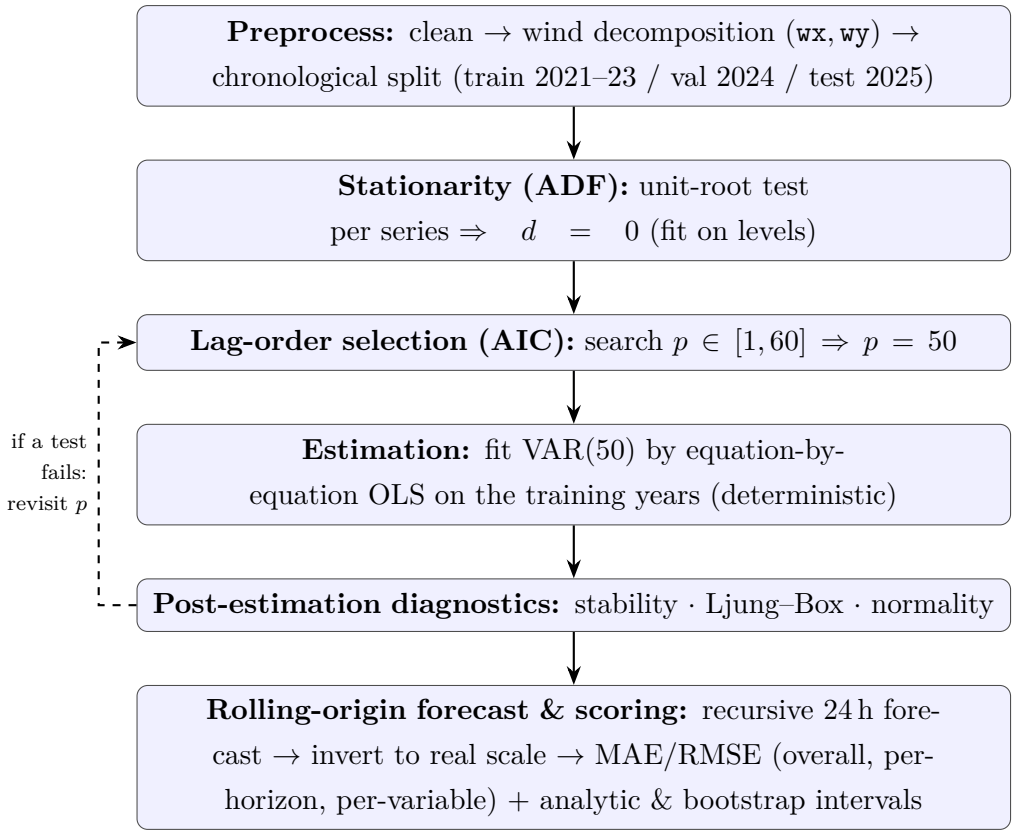


Figure 5.11: VAR modelling pipeline: from raw hourly data to the rolling-origin forecast and its diagnostics. The dashed arrow is the specification-check feedback — re-select the lag order if residual autocorrelation remains.

Algorithm 3 VAR fitting and rolling-origin evaluation

Require: Hourly series; targets k=7; $p_{\max}$=60; horizon H=24

Preprocess: clean, wind decomposition (wx, wy); chronological split
ADF per series on train ⇒ all stationary ⇒ $d = 0$ (levels)
Select p by AIC over $[1, p_{\max}]$ ⇒ $p = 50$
Fit VAR(p) by OLS on 2021–2023 ▷ deterministic
Diagnostics: stability, Ljung–Box, normality (Sec. 5.4.5)
for each test origin t (step 24 h) **do**
    Take last p rows; forecast $t{+}1, \ldots, t{+}H$ recursively (Eq. (5.16))
    Invert to real scale; compare with actuals
end for
Aggregate MAE/RMSE overall, per-horizon, per-variable
Intervals: analytic (Eq. (5.18)) + bootstrap (residual resampling)

5.4.5 Post-estimation Diagnostics

Assumptions and their validity. VAR rests on genuine parametric assumptions, which we verify with the tests defined in Section 5.1; the per-variable results are in

Table 5.17.

- **Stationarity (ADF).** All seven series reject the unit-root null ($p \leq 10^{-8}$) $\Rightarrow$ fit on levels ($d = 0$).
- **Stability.** Every companion root lies inside the unit circle (spectral radius < 1), so the fitted VAR(50) is stable and its forecasts cannot diverge.
- **Residual autocorrelation (Ljung–Box, lag 24).** All seven residual series pass ($p > 0.05$); `pressure` is the tightest ($p = 0.073$), consistent with its longer memory. The residuals are therefore effectively white noise — the chosen lag order leaves no exploitable structure.
- **Normality (Jarque–Bera).** All seven reject normality ($p < 10^{-3}$), driven by heavy tails (excess kurtosis 6–18) and mild skew. This does *not* bias the point forecasts, but it makes the *analytic* Gaussian bands of Eq. (5.18) only approximate; we therefore also report a distribution-free **bootstrap** interval that resamples the empirical residuals (Fig. 5.14).

In summary, the load-bearing assumptions (stationarity, stability, white-noise residuals) all hold; only residual normality fails, and only the interval shape — not the point forecasts or the error metrics — depends on it.

Table 5.17: VAR per-variable diagnostics (residuals of the fitted VAR(50), training set).

Variable	**ADF** p	**Ljung–Box** p	**Jarque–Bera** p	**Skew**	**Kurtosis**
`temp`	1.0×10^{-8}	0.999	$< 10^{-3}$	-1.22	13.74
`rh`	2.8×10^{-23}	1.000	$< 10^{-3}$	0.52	8.84
`dew`	7.6×10^{-10}	1.000	$< 10^{-3}$	-0.45	17.82
`pressure`	2.1×10^{-10}	0.073	$< 10^{-3}$	0.09	5.85
`cloud`	2.8×10^{-27}	1.000	$< 10^{-3}$	-0.13	8.13
`wx`	1.1×10^{-24}	1.000	$< 10^{-3}$	0.44	10.26
`wy`	5.5×10^{-26}	1.000	$< 10^{-3}$	0.07	8.49

Stability holds globally (spectral radius < 1) and the joint multivariate normality test is likewise rejected ($p \approx 0$), in line with the per-variable verdicts.

5.4.6 Experimental Results

Evaluated by rolling-origin over 2025 ($N = 364$ origins, real scale; 95% CIs from 1000 origin bootstraps), VAR(50) attains an **overall MAE of** 5.57 and **RMSE of** 12.07 (Table 5.18); being an OLS fit it is exactly deterministic, so no seed spread is reported.

The per-variable breakdown (Table 5.19) mirrors the pattern of every model in this study: `temp`, `dew`, and `pressure` are forecast tightly ($R^2 \geq 0.92$, MAE ≈ 1.1 in native units), while `cloud` is by far the hardest ($R^2 = 0.37$, MAE 22.8%) — the bimodal, weakly-correlated target flagged in the EDA. Error grows smoothly with lead time, from MAE 2.27 at h=1 to 6.81 at h=24 (Fig. 5.12), and the forecasts track the actuals closely for the well-modelled variables (Fig. 5.13). Notably, this linear, fully interpretable model lands within ∼2% MAE of the much more complex XGBoost (overall MAE 5.48, Section 5.5.8) and is essentially tied with the LSTM on per-variable R^2 (e.g. `temp` 0.918 vs. 0.921; `pressure` 0.954 vs. 0.941), while additionally supplying formal diagnostics and analytic uncertainty bands.

Table 5.18: VAR — overall test metrics (real scale, 2025; $N = 364$ origins, 95% bootstrap CI).

Metric	**Value**	**95% bootstrap CI**
Overall MAE	5.57	[5.38, 5.78]
Overall RMSE	12.07	[11.60, 12.57]

Table 5.19: VAR — per-variable error (real scale, horizon-averaged, 2025; R^2 vs. test mean).

	`temp` (°C)	`rh` (%)	`dew` (°C)	`pressure` (hPa)	`cloud` (%)	`wx` (km/h)	`wy` (km/h)
MAE	1.11	5.60	1.10	1.10	22.82	4.22	3.05
RMSE	1.55	7.51	1.59	1.63	30.08	5.74	4.28
R^2	0.92	0.76	0.94	0.95	0.37	0.52	0.29

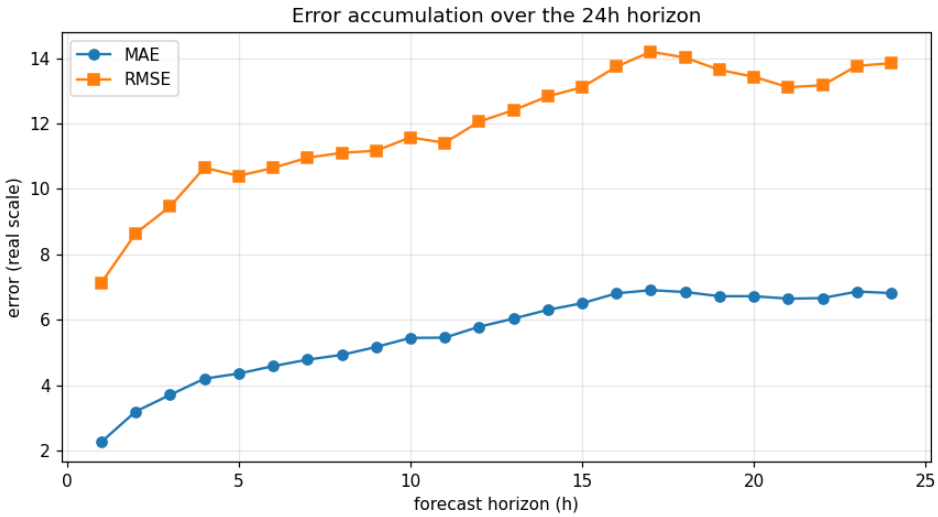


Figure 5.12: VAR: error accumulation over the 24-hour horizon (2025 test).

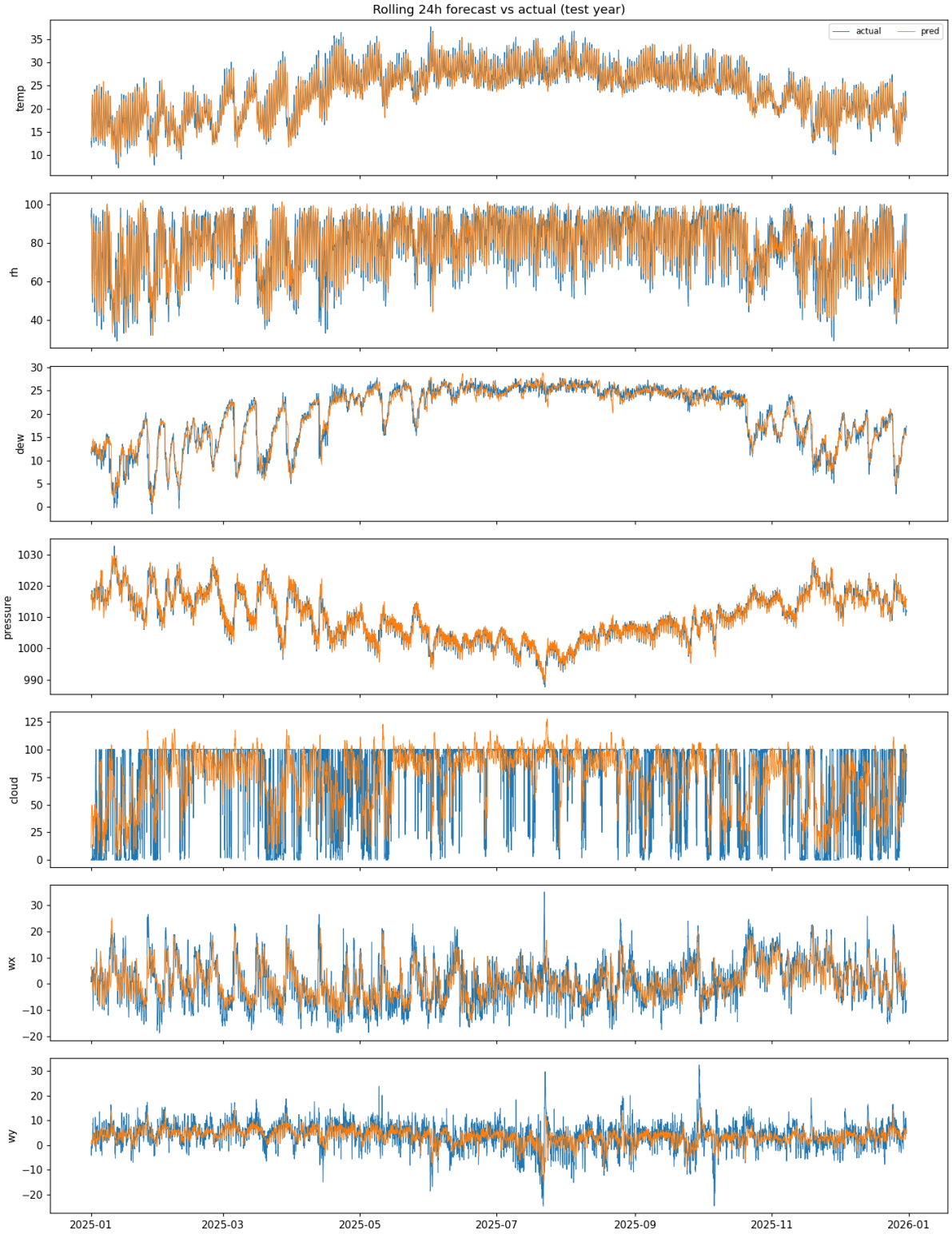


Figure 5.13: VAR: rolling 24-hour forecast vs. actual across the 2025 test year.

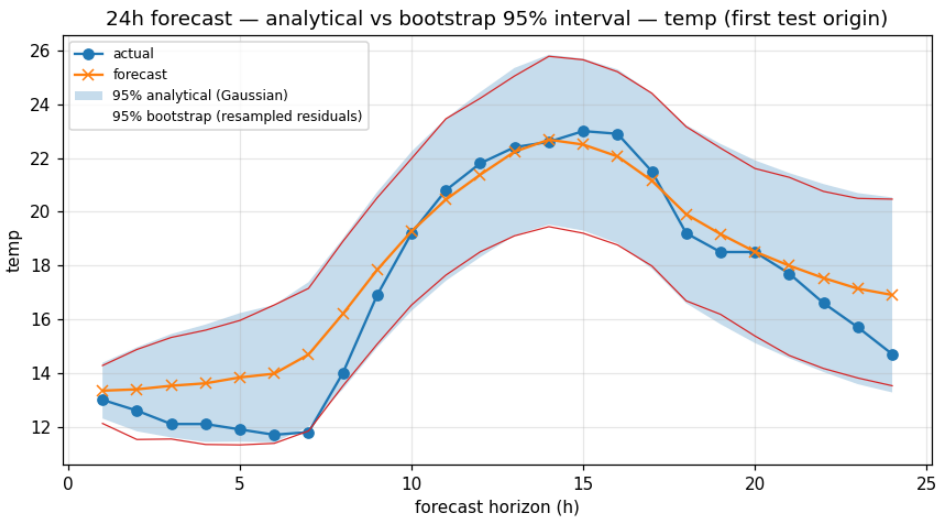


Figure 5.14: VAR: 24-hour `temp` forecast at the first 2025 origin with 95% analytic (Gaussian) and bootstrap (resampled-residual) intervals.

5.5 XGBoost

5.5.1 Overview

XGBoost is a scalable gradient-boosted decision-tree system [2]. In this project the 24-hour-ahead multivariate forecast is produced by **seven multi-output regressors**, one per target variable; each regressor maps the engineered window features to that variable's 24 horizon values. This *direct* (non-recursive) multi-horizon strategy predicts all 24 lead times at once and so avoids the error compounding of recursive one-step feedback.

Why we include it. XGBoost is our *nonlinear machine-learning* reference. It is the strongest off-the-shelf model for tabular, feature-engineered data, captures nonlinear interactions and thresholds that the linear models (VAR, DLinear) cannot, yet trains in seconds on CPU and needs little tuning. It tests how much accuracy comes from nonlinearity once the lag/rolling structure of the series is handed to the model as features — a fairer, cheaper foil to the deep LSTM.

5.5.2 Mathematical Formulation

Gradient boosting builds an additive model of K trees,

$$\hat{y}_i = \sum_{k=1}^{K} f_k(\mathbf{x}_i), \qquad f_k \in \mathcal{F},$$

where $\mathcal{F}$ is the space of regression trees. The regularised objective is

$$\mathcal{L} = \sum_i \ell(y_i, \hat{y}_i) + \sum_k \Omega(f_k), \qquad \Omega(f) = \gamma T + \tfrac{1}{2}\lambda \|\mathbf{w}\|^2, \tag{5.19}$$

with T leaves and leaf weights $\mathbf{w}$. At round t, a second-order Taylor expansion of ℓ around the current prediction gives

$$\mathcal{L}^{(t)} \simeq \sum_i \Big[g_i f_t(\mathbf{x}_i) + \tfrac{1}{2} h_i f_t(\mathbf{x}_i)^2 \Big] + \Omega(f_t), \tag{5.20}$$

with first- and second-order gradients

$$g_i = \partial_{\hat{y}^{(t-1)}} \ell\big(y_i, \hat{y}_i^{(t-1)}\big), \qquad h_i = \partial^2_{\hat{y}^{(t-1)}} \ell\big(y_i, \hat{y}_i^{(t-1)}\big). \tag{5.21}$$

For a fixed structure with leaf instance sets I_j, the optimal leaf weight and structure score are

$$w_j^\star = -\frac{\sum_{i \in I_j} g_i}{\sum_{i \in I_j} h_i + \lambda}, \qquad \tilde{\mathcal{L}} = -\tfrac{1}{2} \sum_{j=1}^{T} \frac{\big(\sum_{i \in I_j} g_i\big)^2}{\sum_{i \in I_j} h_i + \lambda} + \gamma T, \tag{5.22}$$

and splits are chosen greedily to maximise the gain implied by Eq. (5.22). With the squared-error loss $\ell = \frac{1}{2}(y_i - \hat{y}_i)^2$ used here, $g_i = \hat{y}_i - y_i$ and $h_i = 1$.

5.5.3 Feature Engineering

A decision tree splits on individual tabular features and cannot exploit a raw, flattened $120 \times d_{\text{in}}$ window — each of the $\sim$1.3k cells would be a near-duplicate column with little standalone signal. The window is therefore *summarised* into compact per-channel descriptors that expose the level, recent change, volatility, and range of each channel to the splitting criterion. With 7 weather channels the construction yields $7 \times 20 + 4 = \mathbf{144}$ features. Crucially, *every* feature uses only history up to the origin e (lags use $e - L$; rolling statistics use the window $[e - w + 1,\, e]$), so it shares the exact no-leakage horizon of the raw window. In practice the rolling operators are computed once over the whole series and then sampled at the origins, which is both fast and leakage-safe.

Lag Features

For channel c and lag set $\mathcal{S}_{\text{lag}} = \{0, 1, 2, 3, 6, 12, 24, 48, 72, 119\}$,

$$\text{lag}_L^{(c)}(e) = x_{e-L}^{(c)}, \qquad L \in \mathcal{S}_{\text{lag}}.$$

The short lags (0–6 h) capture the immediate trajectory, 12/24 h capture the diurnal cycle, and 48–119 h capture multi-day persistence.

Rolling Mean Features

For windows $w \in \{3, 6, 12, 24, 72\}$,

$$\text{mean}_w^{(c)}(e) = \frac{1}{w} \sum_{i=0}^{w-1} x_{e-i}^{(c)},$$

smoothing the channel at intra-day to multi-day scales to give the tree a denoised level signal.

Rolling Standard-Deviation Features

For windows $w \in \{6, 24, 120\}$,

$$\text{std}_w^{(c)}(e) = \sqrt{\frac{1}{w-1} \sum_{i=0}^{w-1} \left(x_{e-i}^{(c)} - \text{mean}_w^{(c)}(e)\right)^2},$$

summarising short- and long-range volatility (how unsettled recent weather has been), which lets the model sharpen or widen its response.

Rolling Min/Max (Range) Features

A 24-hour minimum and maximum per weather channel describe the recent daily range,

$$\min_{24}^{(c)}(e) = \min_{0 \le i < 24} x_{e-i}^{(c)}, \qquad \max_{24}^{(c)}(e) = \max_{0 \le i < 24} x_{e-i}^{(c)}.$$

Time-based Features

The four cyclical encodings ($\text{hour}_{\sin}, \text{hour}_{\cos}, \text{doy}_{\sin}, \text{doy}_{\cos}$) are taken at the origin so the model knows "what time it is now". The EDA (Chapter 3) showed strong diurnal and annual cycles, which these encodings make directly available to the trees.

5.5.4 Training Procedure

One regressor per target. Seven independent regressors are trained, one per target; each predicts that variable's 24 horizons simultaneously using the default `one_output_per_tree` multi-output strategy. The vector-leaf `multi_output_tree` mode was tested and rejected — on this feature set it was roughly an order of magnitude slower for no accuracy gain.

Data usage and windows. Models are fit on the 2021–2023 training windows with a 6-hour origin step: consecutive hourly windows are near-identical, so a 6-hour step retains ∼4.4k diverse windows at a fraction of the cost, with negligible accuracy loss. The 24-hour evaluation step is unaffected.

Validation and early stopping. The number of boosting rounds is the *only* quantity chosen from data: training monitors the 2024 validation RMSE and stops when it has not improved for 25 rounds (cap 400). All other hyperparameters (Table 5.20) are fixed *a priori* — there is no grid search — and the 2025 test set is never seen during training.

Determinism. Because subsampling makes the learner stochastic, three seeds $\{0, 1, 2\}$ are trained and reported as a seed ensemble together with the per-seed standard deviation.

Table 5.20: XGBoost hyperparameters (fixed a priori; trees chosen by early stopping).

Parameter	Value
objective	`reg:squarederror`
tree method	`hist`
n_estimators (cap)	400
early stopping rounds	25
learning rate	0.08
max depth	6
subsample	0.8
colsample_bytree	0.6
min_child_weight	5.0
reg_lambda	1.0
train origin step	6 h
seeds	{0, 1, 2}

5.5.5 Forecasting Procedure

1. **Preprocessing:** clean, decompose wind, add time encodings, and scale with the train-fit `StandardScaler`.
2. **Feature generation:** encode the 120-hour window ending at the origin into the 144 features of Section 5.5.3.
3. **Prediction:** each regressor outputs its 24 scaled horizon values; stacking the seven gives a 24×7 block (averaged over seeds).
4. **Post-processing:** invert the scaler to the real scale before scoring; optional empirical residual intervals quantify uncertainty.

Figure 5.15 sketches this structure: the lookback window is encoded into engineered features that feed seven independent gradient-boosted ensembles (one per target), each an additive sum of K regression trees, whose 24-step outputs are stacked into the forecast block.

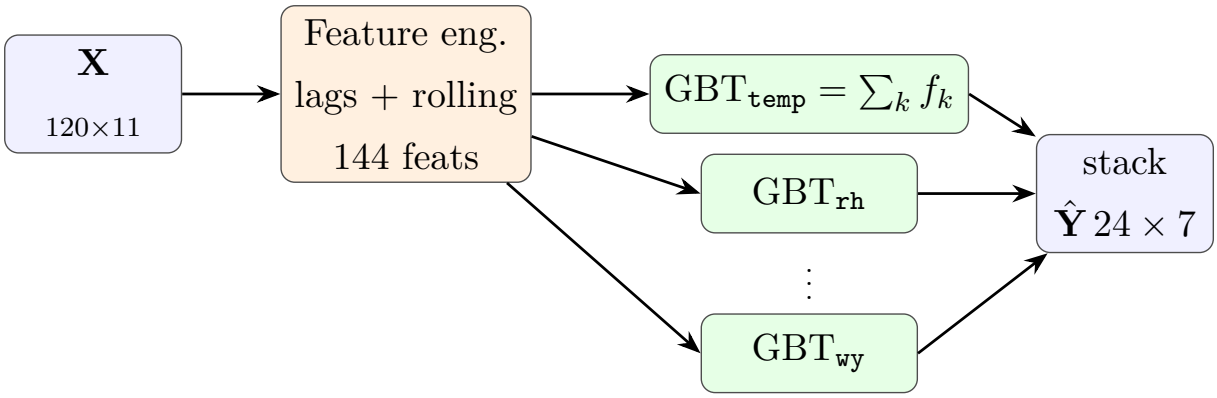


Figure 5.15: XGBoost prediction structure. The 120×11 window is encoded into 144 engineered features feeding seven independent gradient-boosted tree ensembles (one per target), each an additive model $\sum_k f_k$; their 24-step outputs are stacked into the 24×7 forecast block.

5.5.6 Algorithm

Algorithm 4 XGBoost direct multi-horizon forecasting (training + rolling evaluation)

Require: Hourly series; lookback $L = 120$; horizon $H = 24$; targets $V = 7$; seeds

Preprocess: clean, wind decomposition, time encodings
Fit `StandardScaler` on train; transform all splits
Build engineered windows $(\mathbf{X}, \mathbf{Y})$ for train/val/test
for each seed **do**
    for each target variable $v \in \{1, \ldots, V\}$ **do**
        Train multi-output XGBoost $\mathbf{X}_{\text{tr}} \to \mathbf{Y}_{\text{tr}}[:, :, v]$
        Early-stop on $(\mathbf{X}_{\text{val}}, \mathbf{Y}_{\text{val}}[:, :, v])$
    end for
end for
for each test origin t (step 24 h) **do** ▷ rolling-origin evaluation
    Encode window ending at t; predict $\hat{\mathbf{Y}} \in \mathbb{R}^{H \times V}$
    Inverse-transform $\hat{\mathbf{Y}}$ to real scale; compare to actuals
end for
Aggregate MAE/RMSE overall, per-horizon, per-variable; bootstrap CIs

5.5.7 Post-estimation Diagnostics

Assumptions and their validity. As a gradient-boosted tree ensemble, XGBoost makes *no* parametric assumptions — no stationarity, no linearity, and no residual-distribution requirement. It rests on only two working assumptions:

- **The engineered lag/rolling features expose the temporal dependence** the trees then split on. *Satisfied* — the EDA's autocorrelation and lagged cross-correlation

structure (Section 3.6) is exactly what those features encode, and the residual Ljung–Box below confirms little is left over (6/7 pass; only `wx` retains structure — headroom, not invalidity).

- **The 2025 test distribution resembles the 2021–24 training distribution** (no regime shift). *Reasonable* — a stable five-year climate with no trend in the rolling statistics (Fig. 3.9).

Gaussian residuals are *not* assumed, so the non-normal residuals below are handled with empirical intervals. Adequacy is otherwise established through held-out behaviour — convergence of the validation learning curves, residual whiteness, and residual normality on the one-step validation residuals (seed 0 shown; the other seeds agree).

Table 5.21: XGBoost post-estimation diagnostics (1-step validation residuals, seed 0).

Target	best_iter	val-RMSE	Ljung–Box p	Normality
`temp`	117	0.260	0.390 (OK)	non-normal
`rh`	110	0.481	0.417 (OK)	non-normal
`dew`	128	0.243	0.320 (OK)	non-normal
`pressure`	78	0.237	0.319 (OK)	non-normal
`cloud`	47	0.789	0.854 (OK)	non-normal
`wx`	64	0.669	5.1×10^{-6} (autocorr.)	non-normal
`wy`	59	0.900	0.241 (OK)	non-normal

Convergence. Every target reaches a finite early-stopped validation-RMSE minimum (best iteration ≤ 128 of 400), so no series diverges. **Residual autocorrelation.** Ljung–Box passes for six of seven targets; only `wx` retains structure at lag 24 — expected headroom for a direct multi-horizon learner, not invalid forecasts. **Normality** is rejected for all seven (heavy-tailed, bounded targets), so analytical Gaussian bands are replaced by empirical residual intervals, whose 95% half-width grows monotonically over the horizon (e.g. `temp` 0.80 $\rightarrow$ 2.73, validation coverage 0.945).

5.5.8 Experimental Results

Evaluated by rolling-origin over the 2025 test year ($N = 365$ origins, real scale), the seed-ensemble XGBoost attains an **overall MAE of** 5.48 (95% bootstrap CI $[5.28, 5.69]$) and **RMSE of** 11.93 ($[11.44, 12.40]$); the seed standard deviation is only 0.016 (MAE), i.e. the model is effectively deterministic (Table 5.22). The overall figure mixes units, so the per-variable breakdown (Table 5.23) is more informative: temperature, dew point, and pressure are forecast tightly (MAE $\approx$ 1 in their native units), whereas `cloud` is by far the hardest (MAE 22.4%) — consistent with its weak correlations and bimodal

distribution seen in the EDA (Chapter 3). Error grows smoothly with lead time, from MAE 2.26 at h=1 to 6.79 at h=24 before plateauing (Fig. 5.16); the validation learning curves converge without divergence (Fig. 5.17); and the forecasts track the actuals closely for the well-modelled variables (Fig. 5.18).

Table 5.22: XGBoost — overall test metrics (real scale, 2025; seed ensemble $\{0, 1, 2\}$).

Metric	**Value**	**95% bootstrap CI**	**seed std**
Overall MAE	5.48	[5.28, 5.69]	0.016
Overall RMSE	11.93	[11.44, 12.40]	0.012

Table 5.23: XGBoost — per-variable error (real scale, horizon-averaged, 2025).

	`temp` (°C)	`rh` (%)	`dew` (°C)	`pressure` (hPa)	`cloud` (%)	`wx` (km/h)	`wy` (km/h)
MAE	1.06	5.43	1.08	1.24	22.42	4.15	2.99
RMSE	1.49	7.45	1.63	1.80	29.73	5.67	4.17

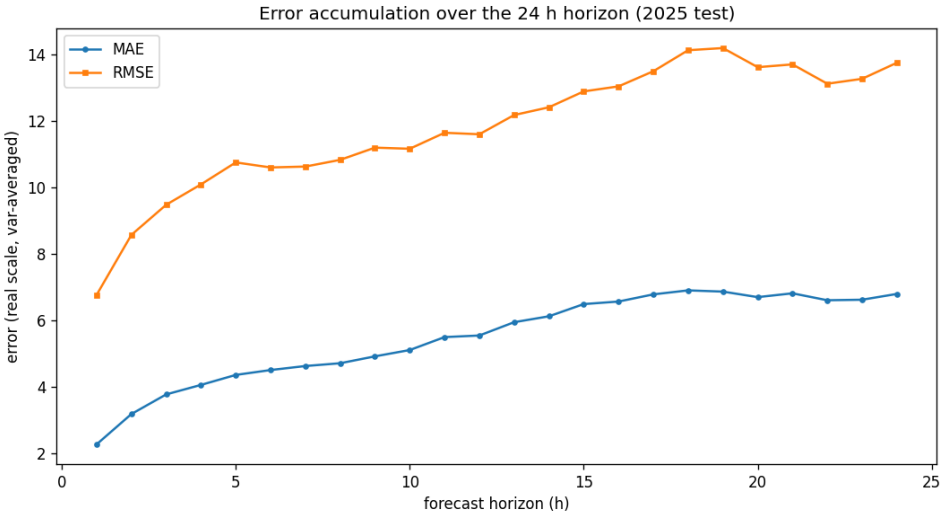


Figure 5.16: XGBoost: error accumulation over the 24-hour horizon (2025 test).

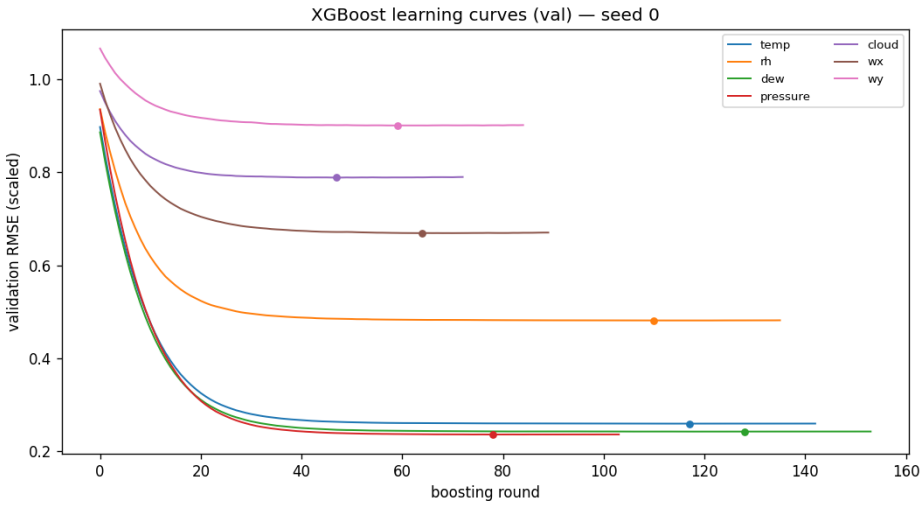


Figure 5.17: XGBoost: validation-RMSE learning curves (seed 0), one curve per target.

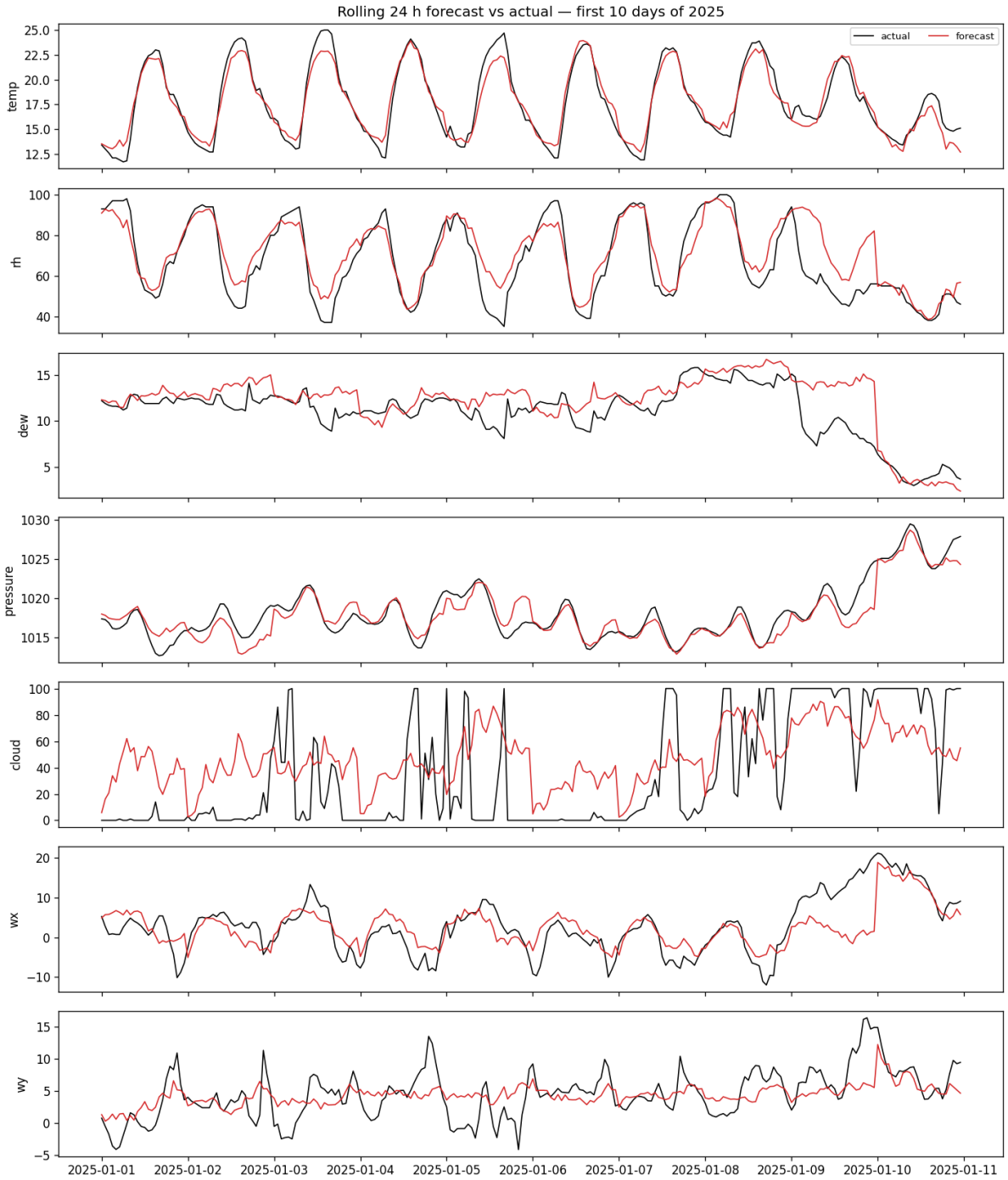


Figure 5.18: XGBoost: rolling 24-hour forecast vs. actual, first 10 days of 2025.

5.6 LSTM

5.6.1 Overview

The Long Short-Term Memory network is a recurrent architecture that learns long-range temporal dependencies via gated memory cells [6]. An encoder LSTM consumes the 120-hour input window and its final hidden state is mapped (directly or through a decoder)

to the 24×7 forecast block. It models nonlinear temporal dependencies without hand-crafted lags, but is more data- and compute-hungry, sensitive to hyperparameters, and less interpretable than the other models.

Why we include it. LSTM is the *deep-learning* representative — the upper end of the capacity spectrum. Unlike XGBoost it learns its own temporal features directly from the raw 120-hour window rather than from hand-engineered lags, so it tests whether end-to-end sequence learning buys accuracy beyond the simpler models. It is the natural ceiling against which the cost of the statistical and tree-based methods is judged.

5.6.2 Mathematical Formulation

At each temporal step $t \in [1, 120]$, the LSTM cell processes the input vector $x_t \in \mathbb{R}^{12\times1}$ and the previous hidden state $h_{t-1} \in \mathbb{R}^{128\times1}$ (given 128 hidden units). To optimize GPU-accelerated tensor computations, the hidden state and input vector are concatenated into a unified vector $[h_{t-1}, x_t] \in \mathbb{R}^{140\times1}$.

The internal mechanics are governed by four primary equations that compute the activations of the forget gate (f_t), input gate (i_t), candidate cell state ($\tilde{c}_t$), and output gate (o_t):

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f) \tag{5.23}$$

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i) \tag{5.24}$$

$$\tilde{c}_t = \tanh(W_c[h_{t-1}, x_t] + b_c) \tag{5.25}$$

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o) \tag{5.26}$$

where:

- $W_f, W_i, W_c, W_o \in \mathbb{R}^{128\times140}$ represent the learnable weight matrices (equivalent to the partitioned matrices $W_{xh} \in \mathbb{R}^{128\times12}$ and $W_{hh} \in \mathbb{R}^{128\times128}$). These matrices act as static, learned filters that remain constant during inference.
- $b_f, b_i, b_c, b_o \in \mathbb{R}^{128\times1}$ are the learnable bias vectors.
- $\sigma(z) = \frac{1}{1+e^{-z}}$ is the logistic sigmoid function, mapping any real input strictly to the range $(0, 1)$ to serve as scaling coefficients.
- $\tanh(z) = \frac{e^z-e^{-z}}{e^z+e^{-z}}$ is the hyperbolic tangent function, mapping values to the range $(-1, 1)$ to establish zero-centered candidate signals.

Once the gates are computed, the cell state $c_t \in \mathbb{R}^{128\times1}$ (carrying long-term sequence trends and contextual variations across the 120-hour window) and the hidden

state $h_t \in \mathbb{R}^{128\times 1}$ (representing localized working memory and imminent hourly dynamics) are updated via:

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t \tag{5.27}$$

$$h_t = o_t \odot \tanh(c_t) \tag{5.28}$$

Here, $\odot$ denotes the Hadamard (element-wise) product. Rather than acting as stochastic filters, the gating outputs represent precise, deterministic scaling coefficients determined by the inputs and the learned weights.

Dual-Dense Layer Pipeline: The network directs the final hidden state summary h_{120} through two distinct fully connected (Dense) layers:

$$h_{dense} = \text{ReLU}(W_{d1}h_{120} + b_{d1}) \tag{5.29}$$

$$\hat{y}_{flat} = W_{d2}h_{dense} + b_{d2} \tag{5.30}$$

- *Dense Layer 1 (Hidden Layer with ReLU):* The Rectified Linear Unit ($\text{ReLU}(z) = \max(0, z)$) is utilized to introduce non-linearity. By clipping negative activations to zero, ReLU encourages sparse representations—allowing specific neurons to specialize in distinct meteorological features—while ensuring robust gradient propagation (preventing vanishing gradients since the gradient of ReLU for positive inputs is a constant 1).

- *Dense Layer 2 (Output Layer with Linear Activation):* To ensure the model can forecast physical values below and above the mean, negative values must not be clipped. Hence, this final layer uses a linear activation function. Because W_{d2} is mathematically unbounded, it acts as a learned linear amplifier, magnifying the signals from the previous layers to project them accurately into any extreme range.

5.6.3 Architecture and Regularization

Unlike tree-based models that require manual feature extraction (e.g., rolling statistics), the LSTM directly digests the raw flattened 120×12 tensor. The network architecture consists of:

1. **LSTM Layer:** Processes the temporal sequence. The parameter `return_sequences=False` ensures only the final hidden state is passed forward.

2. **Dropout Layer:** Randomly zeroes out a fraction of connections during training to break co-adaptation among neurons, heavily penalizing overfitting.

3. **Hidden Dense Layer:** Uses a Rectified Linear Unit (ReLU) activation to learn non-linear combinations of the LSTM's extracted features.

4. **Output Dense & Reshape:** A linear layer producing exactly $24 \times 7 = 168$ nodes, formatted to match the required protocol output.

Figure 5.19 summarises this data path from the input window to the 24×7 forecast block.

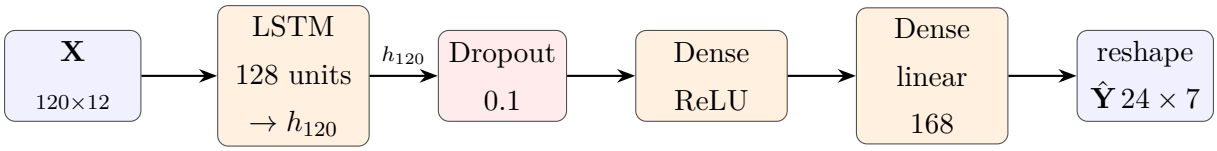


Figure 5.19: LSTM architecture. An encoder LSTM consumes the 120×12 window and emits its final hidden state h_{120}; dropout regularises it, and two dense layers (ReLU then linear) project to the 168-node output reshaped to the 24×7 forecast block.

5.6.4 Training Procedure

Hyperparameter Search Space Strategy: Neural networks are highly sensitive to architectural configurations. To prevent data leakage, a systematic grid search is executed strictly on the 2024 validation set. The search space is carefully bounded to balance model capacity and computational efficiency:

- *LSTM Units* $\{64, 128\}$*:* Controls the representational capacity of the network. While 128 units capture more complex meteorological non-linearities, 64 units act as a structural regularizer against overfitting.

- *Dropout Rate* $\{0.1, 0.2\}$*:* Determines the noise injected during training to mitigate neuronal co-adaptation.

- *Batch Size* $\{64, 128\}$*:* A smaller batch provides a noisier, more frequent gradient update to escape local minima, whereas a larger batch stabilizes the gradient descent trajectory.

- *Initial Learning Rate* $\{10^{-3}, 5 \times 10^{-4}\}$*:* Establishes the momentum of the Adam optimizer during the initial epochs.

Adaptive Optimization (ReduceLROnPlateau): Employing a static learning rate often leads to optimization plateaus or divergent oscillations (overshooting) near the global minimum. To address this, an adaptive learning rate schedule is implemented. The network continuously monitors the validation loss (`val_loss`). If no improvement is observed for 2 consecutive epochs (a patience threshold), the learning rate is scaled down

by a factor of 0.5. This dynamic decay allows the model to take aggressive steps initially and fine-tune its weights meticulously as it approaches the optimal solution.

Overfitting Prevention (Early Stopping): Coupled with the adaptive learning rate, an Early Stopping mechanism operates with a higher patience of 5 epochs. This hierarchical callback structure ensures that the model is first given opportunities to recover via learning rate reductions. If stagnation persists globally, training halts exactly when the model's generalization capability peaks, automatically restoring the optimal weights and preventing the memorization of training noise.

Determinism via Seed Ensemble: Weight initialization in deep learning is stochastic. To ensure reproducibility and robust reporting as required by the protocol, the final chosen architecture is trained across three fixed random seeds $(42, 123, 2026)$. The ultimate forecast is the ensemble mean of these three runs.

Table 5.24: LSTM Hyperparameter Configuration and Tuning Details.

Hyperparameter / Setting	**Value**
Architecture & Data Pipeline	
Lookback Window (L)	120 hours
Forecast Horizon (H)	24 hours
LSTM Hidden Units	128
Dropout Rate	0.1
Optimization	
Optimizer	Adam
Initial Learning Rate	0.001
Batch Size	64
Loss Function	Mean Squared Error (MSE)
Callbacks & Ensemble	
ReduceLROnPlateau Factor	0.5 (Patience: 4 epochs)
Early Stopping Patience	12 epochs
Ensemble Random Seeds	42, 123, 2026

5.6.5 Algorithm

Algorithm 5 LSTM Direct Multi-horizon Forecasting (Tuning & Evaluation)

Require: Hourly series; lookback $L = 120$; horizon $H = 24$; targets $V = 7$; seeds $S = \{42, 123, 2026\}$

Preprocess: clean, decompose wind, encode time, scale via StandardScaler
Generate sliding windows (X, Y) for train (2021-2023), val (2024), test (2025)
Phase 1: Hyperparameter Tuning
for each combination of (units, dropout, batch_size, learning_rate) **do**
    Initialize Adam optimizer with *learning_rate*
    Train model on X_{tr}, Y_{tr}, monitor MSE on X_{val}, Y_{val}
    Apply *ReduceLROnPlateau* (patience=2, factor=0.5)
    Apply *EarlyStopping* (patience=3)
end for
Select best hyperparameters P_{best} based on minimum *val_loss*
Phase 2: Ensemble Training & Evaluation
for each seed $s \in S$ **do**
    Set global random seed s
    Initialize LSTM with P_{best}
    Train on X_{tr}, Y_{tr}, early-stop on X_{val}, Y_{val} (patience=5)
    Predict $\hat{Y}^{(s)}_{scaled}$ on X_{test}
    Inverse transform $\hat{Y}^{(s)}_{scaled} \rightarrow \hat{Y}^{(s)}_{real}$
end for
Calculate Ensemble Mean: $\hat{Y}_{ens} = \frac{1}{|S|} \sum \hat{Y}^{(s)}_{real}$
Compute MAE, RMSE (overall, per-horizon, per-variable) on $\hat{Y}_{ens}$ vs Y_{test}

5.6.6 Post-estimation Diagnostics

Assumptions and their validity. The LSTM is non-parametric in distribution: it assumes neither stationarity nor normal residuals. Its working assumptions, and how each holds, are:

- **The 120-h window carries the signal for the 24-h forecast.** *Satisfied* — the short-lag autocorrelation and daily structure in the EDA (Section 3.6) sit well inside a 120-h lookback.
- **The past→future mapping is stable between train and test** (no regime shift). *Reasonable* — the rolling statistics are flat over the five years (Fig. 3.9).
- **Enough data to fit the network.** *Satisfied* — 26,137 training windows for ∼100k

parameters, averaged over three seeds, with a small train–validation gap (no overfitting).

- **Gaussian residuals** (implicit in the MSE objective). *Violated* — residuals are non-normal and, for the smooth thermodynamic variables, mildly autocorrelated (below). This does not bias the point forecasts, but uncertainty is therefore reported with empirical/bootstrap bands rather than Gaussian ones.

Post-estimation diagnostics were conducted on the $h = 1$ forecast residuals to validate the model specification. The Ljung–Box test (defined in Section 5.1, evaluated at lag 24) assesses whether the residuals resemble independent white noise.

The LSTM successfully captured the underlying dynamics for `rh` ($p = 0.0608$), `cloud` ($p = 0.2695$), `wx` ($p = 0.3948$), and `wy` ($p = 0.1264$), all yielding $p > 0.05$ (Pass). This is a significant achievement, demonstrating that the LSTM effectively extracted all structural patterns from highly chaotic variables like wind vectors and cloud cover.

Conversely, variables characterized by high thermodynamic inertia—such as `temp` ($p = 0.0032$), `dew` ($p < 0.0001$), and `pressure` ($p < 0.0001$)—exhibited residual autocorrelation (Fail). This is physically expected: while the model captures macro-level diurnal cycles accurately, micro-level deviations (due to thermal inertia) tend to persist across consecutive hours, causing the residuals to fail the strict independence assumption without degrading the absolute accuracy.

Table 5.25: Ljung-Box Residual Diagnostic Results

Variable	**p-value**	**Result**	**Status**
temp	0.0032	Reject H_0	Autocorrelated
rh	0.0608	Fail to Reject H_0	White Noise
dew	0.0000	Reject H_0	Autocorrelated
pressure	0.0000	Reject H_0	Autocorrelated
cloud	0.2695	Fail to Reject H_0	White Noise
wx	0.3948	Fail to Reject H_0	White Noise
wy	0.1264	Fail to Reject H_0	White Noise

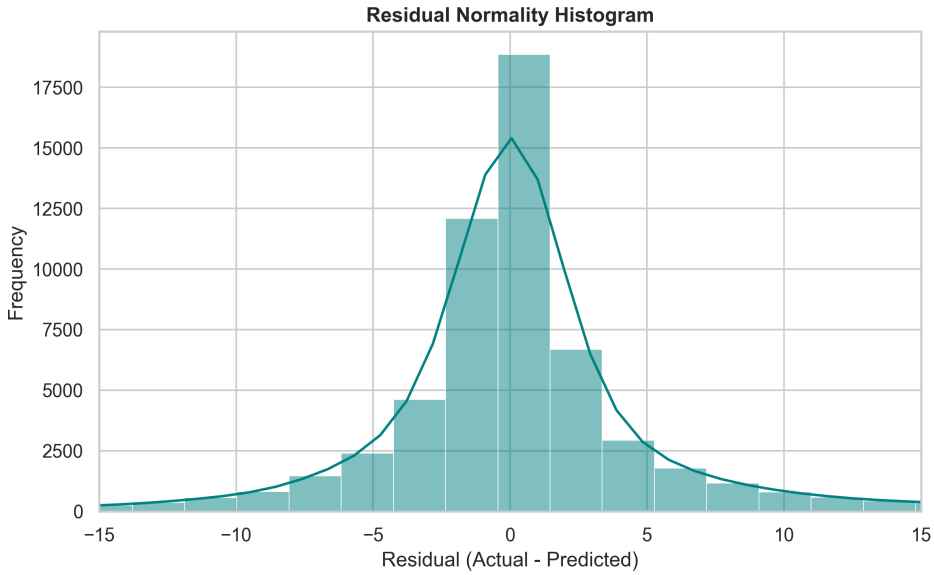


Figure 5.20: LSTM: pooled one-step (h=1) residual distribution. The histogram is sharply peaked at zero with long tails (leptokurtic / heavy-tailed) relative to the Gaussian-like reference curve, departing from normality and consistent with the Jarque–Bera rejection seen across the models.

5.6.7 Experimental Results

Evaluated via the rolling-origin method over the 2025 test set, the tuned LSTM Adaptive LR ensemble demonstrates outstanding predictive capability. Table 5.26 summarizes the overall performance, computed with 1000-iteration Bootstrap 95% Confidence Intervals.

Table 5.26: LSTM Overall Test Metrics (Real Scale, 2025; Ensemble Mean).

Metric	Value [95% CI]
Overall MAE	5.387 [5.157, 5.604]
Overall RMSE	12.010 [11.435, 12.543]
Overall R^2 Score	0.9988

Table 5.27: Per-Variable Metrics with Bootstrap 95% CI on MAE and RMSE.

Variable	**MAE**	**RMSE**	R^2 **Score**
Temperature (`temp`)	1.109 [1.051, 1.173]	1.515 [1.416, 1.623]	0.9207
Relative Humidity (`rh`)	5.649 [5.359, 5.924]	7.580 [7.164, 7.990]	0.7519
Dew Point (`dew`)	1.062 [0.982, 1.143]	1.551 [1.399, 1.706]	0.9390
Pressure (`pressure`)	1.334 [1.244, 1.430]	1.832 [1.677, 1.963]	0.9411
Cloud Cover (`cloud`)	21.415 [19.961, 22.800]	29.918 [28.386, 31.403]	0.3539
Wind Vector X (`wx`)	4.181 [3.942, 4.413]	5.668 [5.256, 6.048]	0.5352
Wind Vector Y (`wy`)	2.959 [2.787, 3.143]	4.116 [3.802, 4.459]	0.3452

Variable-Level & Horizon Analysis: As detailed in Table 5.27, the model achieves an exceptionally high R^2 score (> 0.92) for core thermodynamic variables (`temp`, `dew`, `pressure`), indicating that the LSTM deeply understands diurnal temperature cycles. Figure 5.21 maps the error trajectory over the 24-hour future window, demonstrating that the error accumulation is stable and does not exponentially compound, proving the efficacy of the Direct Multi-step architecture. Figure 5.22 visually confirms the network's ability to smoothly trace real-world meteorological fluctuations without outputting flat averages.

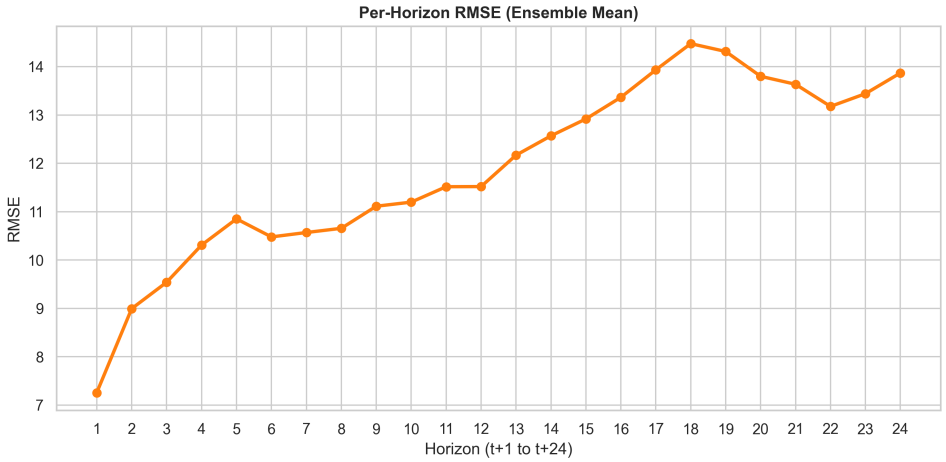


Figure 5.21: LSTM: Error accumulation over the 24-hour horizon (Ensemble Mean, 2025 test).

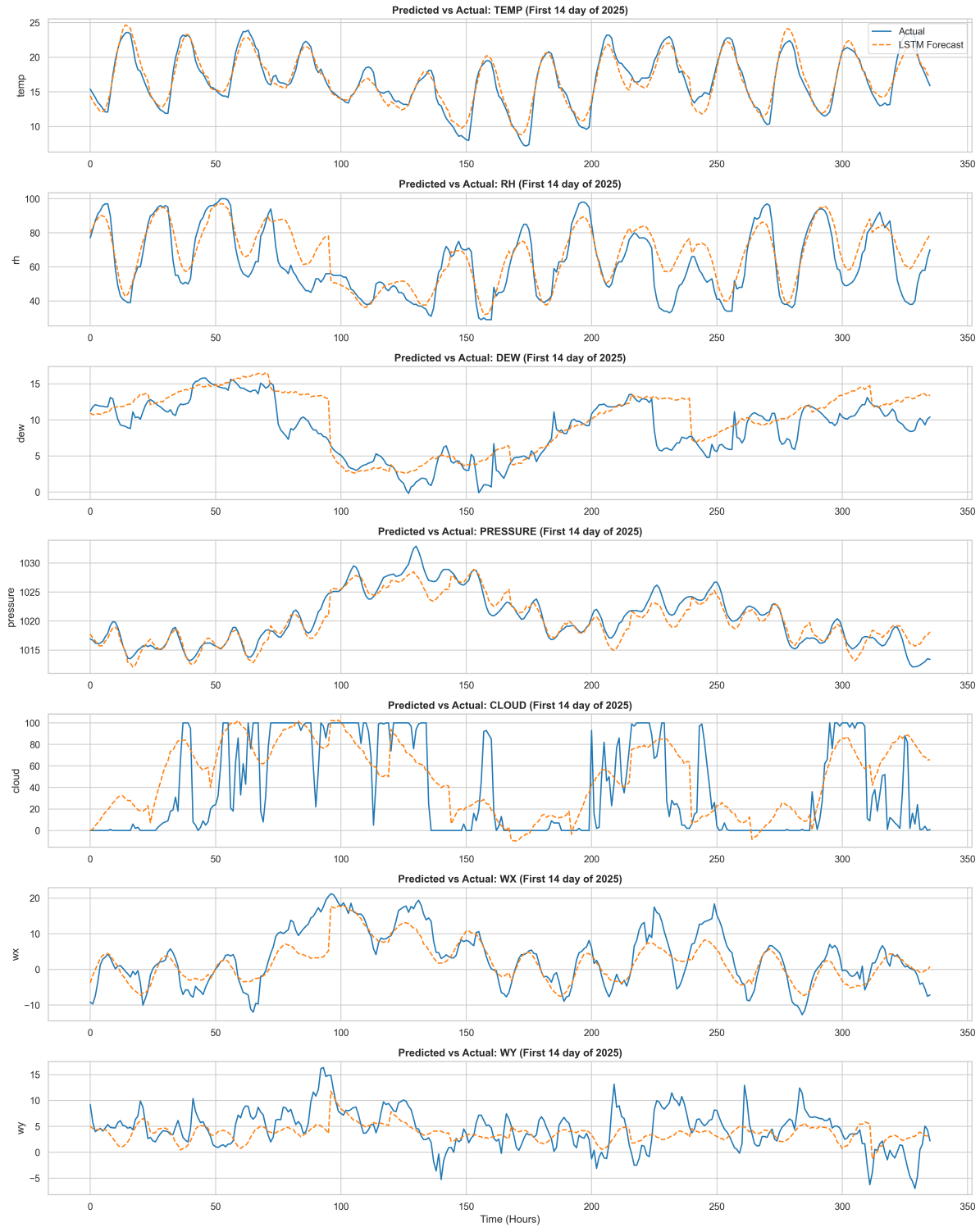


Figure 5.22: LSTM: Rolling 24-hour forecast vs. actual real-world data across all targets.

Chapter 6

Results and Discussion

This chapter brings the five models onto a single scoreboard. All numbers are the 2025 rolling-origin results reported in each model's section (Chapter 5), on the real (denormalised) scale. For the stochastic models the seed-ensemble mean is used, and DLinear is reported as its stronger *individual-weight* variant (DLinear-I).

6.1 Evaluation Protocol

All five models are scored identically with a **rolling-origin** procedure over the 2025 test year. The origin is advanced in non-overlapping 24-hour steps; at each origin t the model conditions on the true history up to t, forecasts $t{+}1, \ldots, t{+}24$, the forecast is inverted to the *real* (denormalised) scale, and is compared with the actual values. This produces $N \approx 365$ forecast blocks, each of shape 24×7 (24 horizons $\times$ 7 variables). Scoring on the real scale keeps errors in physical units (°C, %, hPa, km/h).

Input window. How much past history each model reads at origin t depends on its family. The windowed learners — DLinear, LSTM, and the feature-based XGBoost — use a fixed **120-hour (5-day) lookback** ending at t as their input (the sliding window of Section 4.7). The statistical models condition on history through their fitted lag structure instead: VAR uses its p lags and SARIMA its (seasonal) autoregressive terms, both fed the true series up to t. The forecast *target* — the next 24 hours of all seven variables — is identical for every model, so the comparison is fair regardless of input-window length.

6.2 Evaluation Metrics

Let $\hat{y}_{n,h,v}$ and $y_{n,h,v}$ denote the forecast and the actual value at origin $n \in \{1, \ldots, N\}$, horizon $h \in \{1, \ldots, 24\}$ and variable $v \in \{1, \ldots, 7\}$; the error tensor has $M = N \cdot 24 \cdot 7$ entries.

Mean Absolute Error (MAE, primary).

$$\mathrm{MAE} = \frac{1}{M} \sum_{n,h,v} \left| \hat{y}_{n,h,v} - y_{n,h,v} \right|.$$

Reported in each variable's own unit; robust to outliers.

Root Mean Squared Error (RMSE).

$$\mathrm{RMSE} = \sqrt{\frac{1}{M} \sum_{n,h,v} \left(\hat{y}_{n,h,v} - y_{n,h,v} \right)^2}.$$

Penalises large errors more, so RMSE $\geq$ MAE and the gap widens with the error variance.

Mean Absolute Percentage Error (MAPE).

$$\mathrm{MAPE} = \frac{100\%}{M} \sum_{n,h,v} \left| \frac{\hat{y}_{n,h,v} - y_{n,h,v}}{y_{n,h,v}} \right|.$$

Scale-free, but *undefined when* $y = 0$. Several targets in this project take zero values (precipitation; the wind components `wx`/`wy` when the wind is calm), so MAPE is reported only for strictly positive variables (e.g. `pressure`) and omitted otherwise.

Coefficient of Determination (R^2, per variable).

$$R^2_v = 1 - \frac{\sum_{n,h} (\hat{y}_{n,h,v} - y_{n,h,v})^2}{\sum_{n,h} (y_{n,h,v} - \bar{y}_v)^2},$$

where $\bar{y}_v$ is the test mean of variable v. $R^2 = 1$ is perfect; $R^2 = 0$ ties a constant-mean baseline; $R^2 < 0$ is worse than that baseline.

Reporting breakdowns. The same errors are aggregated three ways: **overall** (average over all n, h, v); **per horizon** (fix h, average over n, v — exposes how error grows with lead time); and **per variable** (fix v, average over n, h — exposes which variables are easy or hard).

Worked example. For `temp` at horizon h=1, suppose three origins give forecasts $\hat{y} = (20, 22, 19)$ and actuals $y = (21, 20, 19)$ (in °C). The absolute errors are $(1, 2, 0)$, so MAE $= \frac{1+2+0}{3} = 1.0$ and RMSE $= \sqrt{\frac{1^2+2^2+0^2}{3}} = \sqrt{5/3} \approx 1.29$. The reported metrics simply extend this same average over all origins, horizons and variables.

Uncertainty. Stochastic models (XGBoost, DLinear, LSTM) are run with ≥ 3 random seeds and reported as a seed mean $\pm$ standard deviation. A 95% **bootstrap** confidence

interval is also obtained by resampling the N forecast origins with replacement and recomputing the overall metric.

6.3 Overall Comparison

Table 6.1 compares the models on the overall metrics. Because the overall figures mix units across variables, the per-variable view (Section 6.4) is the more meaningful comparison.

Table 6.1: Overall test performance (real scale, 2025 rolling-origin).

Model	MAE	RMSE	MAPE (%)	R^2
SARIMA	6.04	12.63	—	—
VAR	5.57	12.07	—	—
DLinear-I	5.64	12.19	—	—
LSTM	**5.39**	12.01	—	—
XGBoost	5.48	**11.93**	—	—

MAE/RMSE are on the real scale; the best value in each column is in bold. MAPE and a single overall R^2 are omitted because the seven targets mix units and several take zero values (calm-wind `wx`/`wy`), which makes a pooled percentage or variance-explained figure ill-defined — the per-variable R^2 in each model's section is the meaningful version. *Seed stability:* the stochastic models are reported as the mean of three seeds; the across-seed standard deviation of the overall MAE is ±0.008 (XGBoost) and ±0.023 (DLinear-I), with the LSTM ensemble of comparable (sub-0.5%) stability. SARIMA and VAR are deterministic. This seed variance is an order of magnitude smaller than the bootstrap confidence intervals (Section 6.2) reported in each model's section and does not affect the ranking.

6.4 Per-Variable Results

Table 6.2 breaks MAE down by target variable.

Table 6.2: Per-variable MAE by model (real scale).

Model	temp	rh	dew	pressure	cloud	wx	wy
SARIMA	1.35	6.14	1.20	1.24	24.45	4.63	3.29
VAR	1.11	5.60	1.10	**1.10**	22.82	4.22	3.05
DLinear-I	1.18	5.64	1.19	1.13	23.00	4.39	3.07
LSTM	1.11	5.65	**1.06**	1.33	**21.42**	4.18	**2.96**
XGBoost	**1.06**	**5.43**	1.08	1.24	22.42	**4.15**	2.99

Per-variable MAE in each target's own unit (°C for `temp`/`dew`, % for `rh`/`cloud`, hPa for `pressure`, km/h for `wx`/`wy`); the best (lowest) value in each column is in bold. Cloud cover is an order of magnitude harder than every other target for all five models. *Seed stability (per variable):* for the stochastic models the across-seed standard deviation of each per-variable MAE is small — for DLinear-I it is $\leq \pm 0.03$ on every target except `cloud` (± 0.11), and XGBoost (overall seed std ± 0.008) and the LSTM ensemble are comparably stable. SARIMA and VAR are deterministic. These spreads are far below the gaps between models and do not change any per-variable winner.

6.5 Per-Horizon Results

Error grows with lead time for every model, from ∼2–3 at h=1 to ∼6.5–7.8 at h=24 (Figure 6.1, Table 6.3). The growth is smooth and saturates rather than diverging. The model ordering, however, *changes* with the horizon. At h=1 the cheap deterministic models are best — SARIMA is lowest (2.23), with XGBoost (2.26) and VAR (2.27) just behind — while the LSTM is the *worst* short-range forecaster (2.84): the direct multi-horizon network spends its capacity on the harder long leads rather than fitting the near term tightly. From h=4 onward the LSTM takes the lead and holds it — by h=24 it is clearly best (6.55 vs XGBoost 6.79, VAR 6.82), and its curve actually *turns down* after $h \approx 18$. SARIMA inverts: best at h=1 but highest at every horizon beyond the first, as its univariate seasonal memory decays with nothing else to lean on.

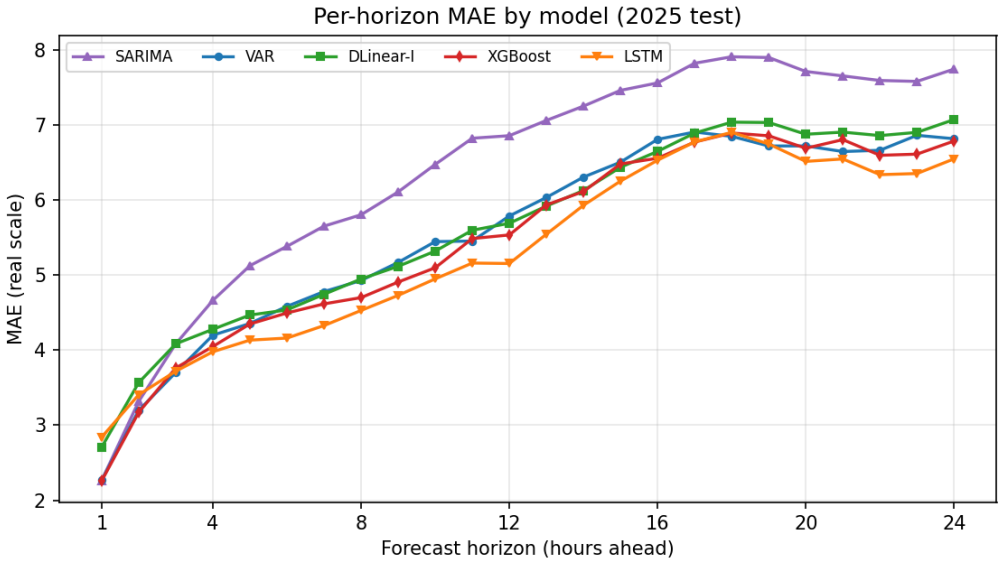


Figure 6.1: Per-horizon MAE by model over the 24-hour horizon (2025 test). The cheap models lead at h=1; the LSTM overtakes from h=4 and is best at the long leads.

Table 6.3: Per-horizon MAE at selected lead times (real scale, 2025). Best per column in bold.

Model	h=1	h=4	h=8	h=12	h=16	h=20	h=24
SARIMA	**2.23**	4.48	5.51	6.40	7.17	7.16	7.27
VAR	2.27	4.20	4.93	5.79	6.81	6.72	6.82
DLinear-I	2.71	4.28	4.95	5.69	6.65	6.88	7.07
XGBoost	2.26	4.05	4.70	5.54	6.56	6.69	6.79
LSTM	2.84	**3.98**	**4.53**	**5.15**	**6.53**	**6.52**	**6.55**

LSTM per-horizon values are from the reproduced three-seed ensemble (overall MAE 5.34, within the bootstrap CI of the 5.39 reported in Section 5.6); the other models are from their own runs.

6.6 Statistical Significance

The tables above rank the models by point estimates. To test whether the differences are *statistically* meaningful — rather than artefacts of a single test year — we run paired forecast-accuracy tests on the per-origin errors. The data for the tests is the set of **aligned loss series**: every model is matched by *(variable, calendar timestamp)* over the 8,617 test hours common to all five pipelines, and at each timestamp t we record its loss $\ell_t^m = |y_t - \hat{y}_t^m|$ (absolute error, matching the MAE headline). For the **overall** test the seven targets are pooled after dividing each error by that target's actual standard deviation, so units are comparable; per-target tests use each target's own unit. Because every model is scored on the *same* timestamps, the pairing removes the shared "hard-day" variance (all models err together when the weather is hard), which makes these tests far more powerful than comparing the marginal bootstrap intervals of Section 6.2.

Diebold–Mariano test (pairwise)

For two models A, B [3] define the loss differential and its mean over the n aligned hours,

$$d_t = \ell_t^A - \ell_t^B, \qquad \bar{d} = \frac{1}{n}\sum_{t=1}^{n} d_t. \tag{6.1}$$

Under $H_0 : \mathbb{E}[d_t] = 0$ the two models are equally accurate. Since the hourly d_t are autocorrelated (consecutive hours share a forecast origin), its variance is estimated with the heteroskedasticity- and autocorrelation-consistent (Newey–West) long-run variance using Bartlett weights up to a one-day lag $L = 23$,

$$\hat{\sigma}^2 = \hat{\gamma}_0 + 2\sum_{k=1}^{L}\Big(1 - \frac{k}{L+1}\Big)\hat{\gamma}_k, \qquad \hat{\gamma}_k = \frac{1}{n}\sum_{t=k+1}^{n}(d_t - \bar{d})(d_{t-k} - \bar{d}). \tag{6.2}$$

The test statistic, with the Harvey–Leybourne–Newbold (HLN) small-sample correction factor c_n, is

$$\mathrm{DM} = c_n \frac{\bar{d}}{\sqrt{\hat{\sigma}^2/n}}, \qquad c_n = \sqrt{\frac{n+1-2L+L(L-1)/n}{n}} \approx 1, \tag{6.3}$$

and is compared to a t_{n-1} distribution. A positive $\bar{d}$ means A has the larger loss (is worse). We run the test for every model against the best model and against the SARIMA baseline; because this is several comparisons at once, the p-values are adjusted by the **Holm** step-down procedure to control the family-wise error.

Model Confidence Set (joint)

The MCS [5] converts the pairwise tests into a single set of "statistically best" models. Starting from the full set $\mathcal{M}$ of five models, it iterates:

1. compute the range equivalence statistic $T_{\mathcal{M}} = \max_{i,j\in\mathcal{M}} |t_{ij}|$, where $t_{ij} = \bar{d}_{ij}/\widehat{\mathrm{sd}}(\bar{d}_{ij})$ is the standardised mean loss difference between models i and j;
2. obtain its null distribution by a **stationary block bootstrap** (5000 resamples, expected block length 24 h, which preserves the daily dependence of the loss series), giving a p-value for H_0: "all models in $\mathcal{M}$ are equally accurate";
3. if H_0 is rejected at size $\alpha = 0.10$, eliminate the single worst model (largest standardised excess loss) and repeat; otherwise stop.

The surviving set is the 90% MCS — the models that cannot be separated from the best. Each model also gets an **MCS p-value** (the level at which it first enters the set): a value above 0.10 means "in the 90% set".

Table 6.4: Forecast-accuracy significance on the overall (unit-standardised) loss. DM statistics are paired vs. the best model (LSTM) and vs. the univariate baseline (SARIMA); p-values are Holm-corrected for multiple comparisons. The 90% Model Confidence Set contains every model not significantly worse than the best.

Model	**Std. loss** (lower better)	**DM vs. best** (Holm p)	**DM vs. SARIMA** (Holm p)	**MCS p**	**In 90% MCS**
LSTM	**0.365**	—	$< 10^{-20}$	1.00	✓
XGBoost	0.366	0.64	$< 10^{-35}$	0.64	✓
VAR	0.372	0.076	$< 10^{-21}$	0.10	✓
DLinear	0.379	1.7×10^{-4}	$< 10^{-22}$	0.00	—
SARIMA	0.409	$< 10^{-20}$	—	0.00	—

Findings. Three conclusions are statistically robust (Table 6.4):

1. **Every learner beats the univariate baseline.** VAR, DLinear, XGBoost and LSTM are all significantly more accurate than SARIMA ($p < 10^{-21}$ in every case). The value of modelling cross-variable structure is therefore not a numerical coincidence — it is decisive.

2. **The best models are a three-way tie.** The 90% MCS is $\{\text{LSTM}, \text{XGBoost}, \text{VAR}\}$. The LSTM has the lowest loss but is *not* significantly better than XGBoost ($p = 0.64$) or, at the 5% level, VAR ($p = 0.076$): a tuned deep network, a tuned gradient-boosted ensemble, and a closed-form OLS VAR are statistically inseparable on this task.

3. **DLinear is the odd one out among the learners.** Despite a competitive MAE, its errors are *consistently* a little larger than the top three across origins, so the paired test rejects equality ($p = 1.7 \times 10^{-4}$) and the MCS excludes it — a difference the marginal bootstrap intervals were too coarse to resolve.

Per-variable confidence sets. No model dominates target-by-target. The 90% MCS per target is {XGBoost} for `temp` and `rh`, $\{\text{VAR}, \text{DLinear}\}$ for `pressure`, {LSTM} for `cloud` and `wy`, and the wider $\{\text{LSTM}, \text{XGBoost}, \text{VAR}\}$ for `dew` and `wx` — mirroring the split-decision per-variable MAE table (Table 6.2).

Caveat. The pipelines roll on slightly different phases, so a given hour is forecast at a (constant) different lead time by each model. Pooled over the test year every model uses each lead equally often, so the mean loss differential still equals the overall-MAE gap and the test is unbiased; the phase only adds noise that the large sample (8,617 paired hours) absorbs.

6.7 Discussion

Most accurate overall. The two nonlinear learners lead the overall table (Table 6.1): LSTM has the lowest MAE (5.39) and XGBoost the lowest RMSE (11.93). The margin over the linear multivariate baseline is small, however — VAR (5.57) and DLinear-I (5.64) sit within ∼5% of the best. The paired significance tests (Section 6.6) make this precise: the 90% Model Confidence Set is $\{\text{LSTM}, \text{XGBoost}, \text{VAR}\}$, so the closed-form VAR is statistically inseparable from the tuned learners, while the highest-MAE model, SARIMA (6.04), is significantly worse than all four ($p < 10^{-21}$). As the only *univariate* model SARIMA cannot use the strong cross-variable coupling (temperature–dew-point–humidity–pressure) that the EDA identified (Section 3.6), and it pays for it. The headline message is therefore that *multivariate* structure matters more than model nonlinearity on this dataset: a closed-form VAR nearly matches a tuned LSTM.

Hardest variables. No model dominates the per-variable table (Table 6.2); the win is split (XGBoost on `temp`, `rh`, `wx`; LSTM on `dew`, `cloud`, `wy`; VAR on `pressure`). The thermodynamic variables (`temp`, `dew`, `pressure`) are forecast tightly by everyone (MAE $\approx$ 1.1–1.4), because they are smooth and strongly autocorrelated. `Cloud` is an order of magnitude harder (21–27% MAE) for all five models: it is effectively bimodal (clear $\approx$ 0% vs overcast $\approx$ 100%), weakly linearly correlated with the other targets, and its transitions are driven by unobserved sub-grid processes — so a squared-error regressor on the available channels has little to lock onto. The wind components are intermediate, limited by the stochastic, gust-driven behaviour of wind at hourly resolution.

Error vs. horizon. Across all families the error grows smoothly with lead time (Figure 6.1), roughly 2.5–3× from h=1 to h=24, and saturates rather than diverging — consistent with stable, well-specified forecasters. The ranking is *not* fixed, though: at h=1 the cheap models (XGBoost, SARIMA, VAR $\approx$ 2.3) beat the LSTM (2.84), but from h=4 the LSTM overtakes and is clearly best by h=24. The direct multi-horizon learners (LSTM, XGBoost) thus trade a little short-range accuracy for materially better long-range forecasts, whereas SARIMA is highest at every horizon past the first.

Accuracy vs. cost. Accuracy buys little once the model is multivariate, but cost varies enormously. SARIMA and VAR are deterministic and cheap — VAR is a single closed-form OLS fit with no seed and no tuning, yet ties the ML models on several variables. DLinear is the cheapest learner ($\sim$85 s, <6k parameters) and lands mid-pack. XGBoost is the strongest all-round value: modest CPU training, near-best on every breakdown. LSTM delivers the best MAE but at the highest price (GPU training, a hyperparameter grid search, and a three-seed ensemble). For deployment on this dataset the marginal accuracy of the deep model is hard to justify over VAR or XGBoost.

What the diagnostics say. The post-estimation checks (Section 5.1) tell a consistent story across all five models. First, every model is *well-behaved for point forecasting*: VAR is stable (roots inside the unit circle), the learners all converge to a flat validation loss without divergence, and the targets are level-stationary (Section 3.6), so the statistical models fit on levels (d=0). Second, *no model fully whitens its residuals*: Ljung–Box leaves significant autocorrelation in the smooth, long-memory thermodynamic variables (`temp`, `dew`, `pressure`) for essentially every model, while the harder, noisier targets (`cloud`, wind) tend to *pass* — the reverse of the accuracy ranking. This is intuitive: what is easy to predict (temperature) still carries fine structure the model has not captured, whereas what is hard (cloud) is closer to noise once the obvious signal is removed; the leftover autocorrelation is head-room, not a sign of an invalid model. Third, *residual normality is rejected everywhere* — by the statistical and the learned models alike — because the

targets are bounded (cloud) and heavy-tailed (wind gusts). Gaussian analytic intervals are therefore unreliable, which is why every model reports empirical or bootstrap intervals instead. In short, the diagnostics reflect the *data* (bounded, gusty, long-memory) more than any modelling choice, and they are what justify MAE as the primary metric and the resampled uncertainty bands used throughout.

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